RAJAGIRI COLLEGE OF SOCIAL SCIENCES

(AUTONOMOUS)

KALAMASSERY

Department of Computer Science

WPL Project Report

AutoVerse

Done by

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Reg No: 24118004

Under the guidance of

Mr. Ajay Antony Joseph

2024-2028

RAJAGIRI COLLEGE OF SOCIAL SCIENCES

(AUTONOMOUS)

KALAMASSERY

CERTIFICATE

AutoVerse

Certified that this is the Bonafide record of project work done by

Abhishek Subramanian

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During the academic year 2024-2028, in partial fulfillment of requirements for award of the degree,

Bachelor of Computer Science (Hons.) with Data Analytics

of

Rajagiri College of Social Sciences (Autonomous)

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Abhishek Subramanian

ABSTRACT

AutoVerse is a web-based platform designed to simplify the process of exploring, comparing, and renting cars while providing detailed editorial reviews to help users make informed decisions. The system allows users to filter cars by brand, compare multiple vehicles, add favourites to a wishlist, rent cars and access car-specific communities with a structured workflow. The platform manages rentals efficiently, including booking, approval, delivery tracking, and invoicing, ensuring a seamless experience for both users and admins. Built using Laravel 12, PGSQL , and Blade templates, AutoVerse incorporates secure user authentication, admin-controlled content management, image handling, and PDF invoice generation. This platform enhances operational efficiency, user convenience, and decision-making in the car rental and review ecosystem.

LIST OF FIGURES

TABLE OF CONTENTS

1. INTRODUCTION

AutoVerse – A Comprehensive Automotive Review, Comparison and Rental Management Platform, is a comprehensive web-based solution designed to streamline the process of exploring, comparing, and renting cars. Built using Laravel 12 for the backend, Blade templates and Bootstrap 5 for the frontend, and PGSQL for the database, the system allows users to access detailed car information, make informed decisions, and manage rentals efficiently, all through an intuitive and responsive interface.

The platform provides detailed editorial reviews for each car, covering interior, driving experience, safety, and other specifications, along with high-quality images. Users can filter cars by brand, compare multiple vehicles side by side, add favorites to their wishlist, and initiate rental requests. The system ensures secure user authentication and personalized access to rental history, invoices, and profile management.

Admins can manage the entire platform, including adding and updating car details, posting reviews, managing rental requests, and monitoring system activity. The rental workflow includes booking, approval, active rentals, returns, cancellations and automatic PDF invoice generation. By automating these processes, AutoVerse reduces manual errors, improves operational efficiency, and provides a seamless experience for both users and administrators.

With real-time rental tracking, secure data management, and a responsive, user-friendly interface, AutoVerse enhances decision-making for users, simplifies administrative workflows, and fosters an efficient and reliable car rental ecosystem.

2. SYSTEM ANALYSIS

2.1 Module Description

The main modules of **AutoVerse** are as follows:

- Car Management
- Review Management
- Rental Management
- Wishlist Management
- User Management
- Authentication & Access Control
- Invoice & Transaction Management

2.1.1 Car Management

This module allows administrators to add, update, and delete car records. It manages core car details such as brand, model, year, rental price, and availability status.

Car images and related editorial content are also linked through this module, ensuring that all car-related information is centrally maintained and easily accessible.

2.1.2 Review Management

This module handles detailed editorial reviews for each car. It manages:

- Title and introduction
- Interior review
- Driving experience
- Safety insights
- Ratings
- Associated images

It ensures that users can access comprehensive and structured car reviews to make informed decisions.

2.1.3 Rental Management

This module manages the entire rental lifecycle, including:

- Rental booking

- Approval by admin
- Active rental tracking
- Completion and cancellation
- Rental status updates

It calculates total rental cost based on selected dates and maintains a structured rental history for each user.

2.1.4 Wishlist Management

This module allows registered users to:

- Add cars to wishlist
- Remove cars from wishlist
- View saved cars

It enhances personalization by allowing users to bookmark cars for future consideration.

2.1.5 User Management

This module enables the system to manage user accounts, including:

- Registration and login
- Profile updates
- Viewing rental history
- Managing account details

Admins can monitor user activity and ensure proper access control.

2.1.6 Authentication & Access Control

This module ensures secure access to the platform through:

- User authentication
- Role-based access control (Admin / User)
- Middleware protection for restricted routes

It protects sensitive system operations and maintains platform security.

2.1.7 Invoice & Transaction Management

This module handles:

- Rental transaction records
- Invoice generation
- Printable invoice format using browser printing
- Tracking rental payment-related information

It ensures financial records are structured and accessible.

Users are the key factor that make the existence of **AutoVerse** viable. The system is designed to cater to different types of users, each with specific roles and responsibilities. The major user categories are described below:

Admin

- The admin is responsible for managing the platform, including car details, editorial reviews, rental requests, and user accounts.
- Admins control access permissions, ensuring secure login practices and maintaining the integrity of sensitive data.
- They approve or reject rental requests, monitor platform activity, and ensure smooth day-to-day operations.

Registered User

- Registered users can browse cars, read detailed editorial reviews, compare multiple vehicles, and add favourites to their wishlist.
- They can place rental requests, track their rental history, view invoices and access car-specific communities.
- Users are responsible for providing accurate personal information and following the rental terms and conditions.

Guest (Optional)

- Guests can browse available cars and read general reviews but cannot access advanced features such as wishlist management or rental booking.

2.2 Business Rules

Business rules define the operational constraints and policies that guide the functionality of the **AutoVerse** platform. These rules ensure that the system operates consistently and that interactions between users and administrators follow defined guidelines.

User Management

- Users must create an account and log in before accessing personalized features such as wishlists, rentals, and community discussions.
- Each user account is uniquely identified by an email address.
- Users are allowed to update their personal profile information, including their name and email.

Car Management

- Only administrators are authorized to add, edit, or remove car entries from the system.
- Each car record must include essential information such as brand, model, year, and rental price.
- Detailed review content for each car is stored separately and linked to the main car entry.

Rental Management

- A user must be logged in to request a car rental.
- Rental requests must include a **start date, end date, and delivery location**.
- Rental requests initially have a **pending status** and require administrator approval.
- Users can cancel rental requests before the rental becomes active.

Wishlist Management

- Users can add cars to their wishlist for future reference.
- A user cannot add the same car to the wishlist more than once.
- Wishlist items can be removed at any time by the user.

Community Interaction

- Only users who have **completed a rental** are allowed to create community posts about cars.
- Users can reply to existing community posts to share experiences or opinions.
- Community discussions are associated with specific cars.

Data Integrity

- All user, car, rental, and community data are stored securely in the system database.
- The system ensures that relationships between users, cars, rentals, and posts remain consistent through database constraints.

2.3 Analysis of Architecture

2.3.1 Frontend Development

The front-end of **AutoVerse** is the part of the platform that users see, interact with, and experience directly. It is designed to provide a smooth, intuitive and responsive interface for browsing cars, reading reviews, managing wishlists, and handling rentals. In this sense, empathy for the user is as important as technical skill—the interface must communicate clearly and be easy to navigate.

The front-end development of **AutoVerse** uses the following technologies:

- HTML5: All web pages in **AutoVerse** are ultimately rendered as HTML. HTML is the backbone of the interface, defining the structure and content that users interact with, such as car listings, review sections, and forms for rentals and user profiles.
- CSS3 and Bootstrap 5: While HTML defines structure, CSS is used to style and present the interface. Bootstrap 5 is used to create responsive layouts, grids, and components. CSS ensures that the platform is visually appealing, with well-structured content, intuitive navigation, and a consistent look and feel.
- JavaScript: JavaScript is used to add dynamic and interactive elements to the platform. Features such as toggling wishlists, filtering cars, comparing vehicles side by side, and managing modal popups for rental bookings are powered by JavaScript. This allows users to interact with **AutoVerse** in real-time without needing to refresh the page for every action.

Overall, the front-end of **AutoVerse** is built to be user-friendly and visually appealing ensuring a smooth and engaging experience.

2.3.2 Back-End Development

The back-end of **AutoVerse** acts as the core engine that powers the entire platform. While the front-end handles what users see and interact with, the back-end is responsible for business logic, database operations, authentication, rental workflows, and overall system performance.

All critical operations—such as user registration, login authentication, car management, wishlist handling, rental booking, status updates, invoice generation and community access—are processed on the server side. Back-end code runs on the server and communicates with the database to retrieve, store, and update information dynamically. A well-structured back-end ensures that the system is secure, efficient, and reliable.

If the system experiences issues such as slow responses, incorrect data processing, or authentication failures, the root cause is typically related to back-end logic or server-side processing.

Laravel Framework

AutoVerse is developed using **Laravel 12**, a modern PHP framework that follows the MVC (Model-View-Controller) architecture.

Laravel provides:

- Structured routing and controller management
- Middleware for authentication and admin access control
- Eloquent ORM for database interactions
- Secure password hashing and session management
- Migration system for database version control
- Built-in validation mechanisms

The MVC architecture helps separate business logic, presentation, and data handling, making the application more organized and maintainable.

PHP

PHP (Hypertext Preprocessor) is the server-side scripting language used to build **AutoVerse**. PHP executes on the server and generates dynamic content that is sent to the browser as HTML.

In **AutoVerse**, PHP is used to:

- Handle form submissions
- Process rental calculations
- Manage user authentication
- Interact with the PostgreSQL database
- Control business logic and application workflows

Laravel enhances PHP by providing a structured and secure development environment.

PostgreSQL

AutoVerse uses **PostgreSQL** as its relational database management system, managed via pgAdmin 4.

PostgreSQL is a powerful, open-source RDBMS known for:

- Strong data integrity
- Advanced indexing and querying capabilities
- High reliability and scalability

- ACID-compliant transactions

It stores all system data, including:

- User accounts
- Car details and editorial content
- Wishlist records
- Rental transactions
- Rental status lifecycle
- Community posts and replies

PostgreSQL ensures that data is stored securely and retrieved efficiently.

Development Environment

AutoVerse is developed and tested using:

- Laravel's built-in development server (php artisan serve)
- PostgreSQL database via pgAdmin 4
- Modern web browsers for testing and debugging

This setup allows efficient local development without requiring external web server packages such as XAMPP or WAMP.

2.4 Feasibility Analysis

2.4.1 Technical Feasibility

AutoVerse is technically feasible as it uses modern and widely supported technologies. Laravel 12 provides a robust backend framework, Blade templates support dynamic and responsive frontend development, and PGSQL ensures reliable data storage and retrieval. The platform can be deployed on standard web servers, making it accessible on desktop and mobile devices without requiring specialized hardware.

2.4.2 Economic Feasibility

The development of **AutoVerse** is financially feasible. Using open-source technologies such as Laravel, PGSQL, and Bootstrap keeps development and deployment costs low. Web hosting, database management, and storage requirements are minimal, allowing the project to be implemented without significant financial investment. By centralizing car reviews, wishlist, and rental operations, the platform also reduces time and effort for users and administrators, providing indirect cost savings.

2.4.3 Operational Feasibility

The system is operationally feasible as it is designed to simplify and streamline the tasks of both users and administrators. Users can easily browse car reviews, manage wishlists, compare cars, and book rentals through an intuitive interface. Administrators can manage cars, reviews, and rental workflows efficiently, ensuring smooth day-to-day operations.

2.4.4 Behavioural Feasibility

Behavioural feasibility is the evaluation and estimation of user attitude towards the development of the new system. **AutoVerse** provides unique value by combining detailed editorial reviews, wishlist management, and rental functionality in one platform—features that are currently fragmented across multiple websites. This makes the system convenient for users, ensuring positive adoption and engagement. Administrators also benefit from centralized management, enhancing overall satisfaction with the platform.

2.5 Actors and Roles

Actor name : User

Role : The user is the primary actor of the **AutoVerse** platform. Users can browse available cars, read detailed reviews, compare vehicles, add cars to their wishlist, request rentals and access car specific communities once they have completed a rental. Users can also manage their personal profile, track rental requests, and view the status of their orders. The system provides an intuitive interface that allows users to interact with the platform easily and perform all operations related to car browsing and rental services.

Actor Name : Administrator

Role : The administrator is responsible for managing and maintaining the AutoVerse platform. The administrator can add, edit, or delete car records, post and manage car reviews, and monitor rental requests submitted by users. Administrators also approve or reject rental requests, update rental statuses, and ensure that the system data remains accurate and up to date.

3. SYSTEM DESIGN

3.1 Use case model

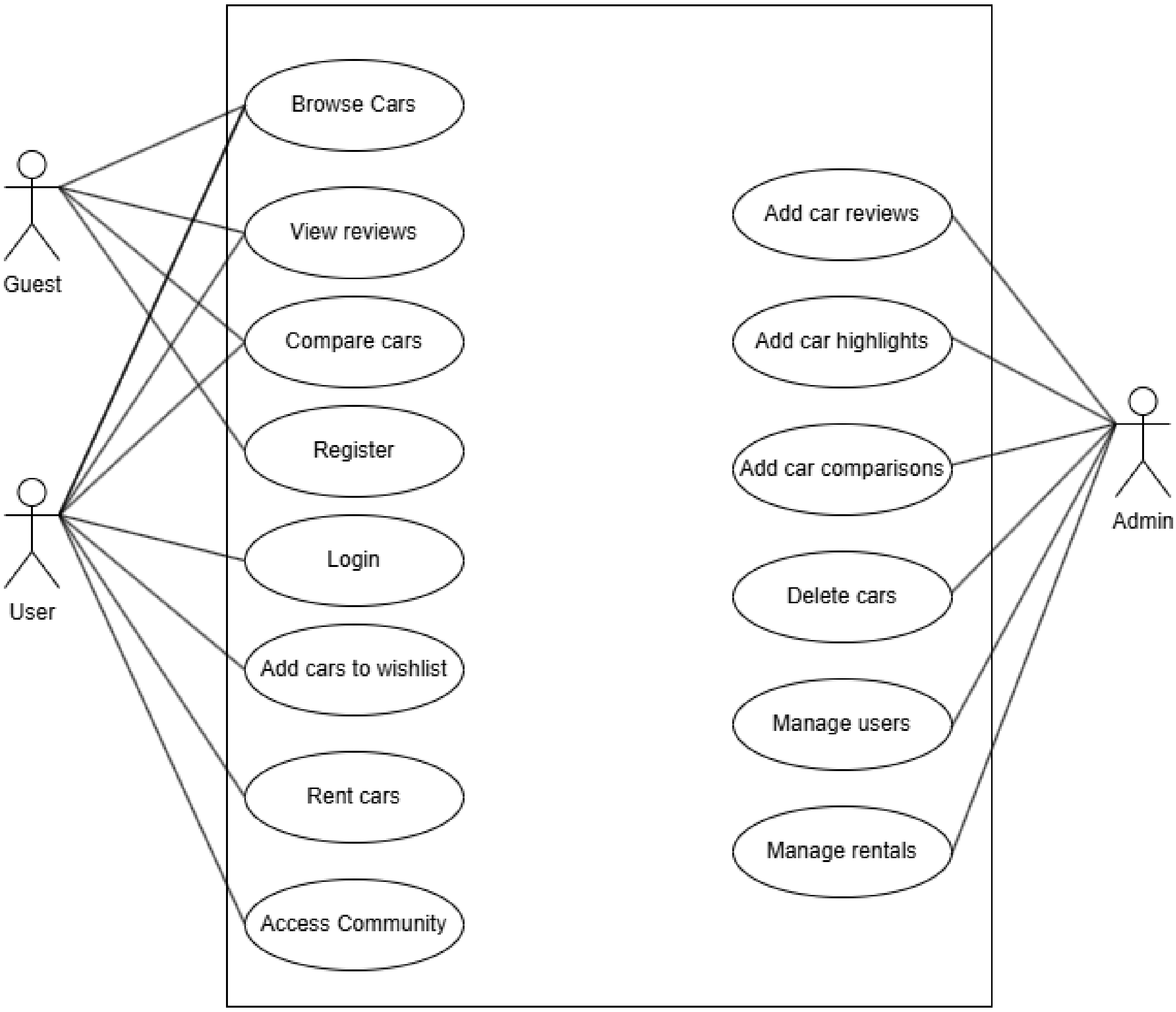


Figure 3.1 : Use case diagram

3.2 Sequence Diagrams

3.2.1 User Login

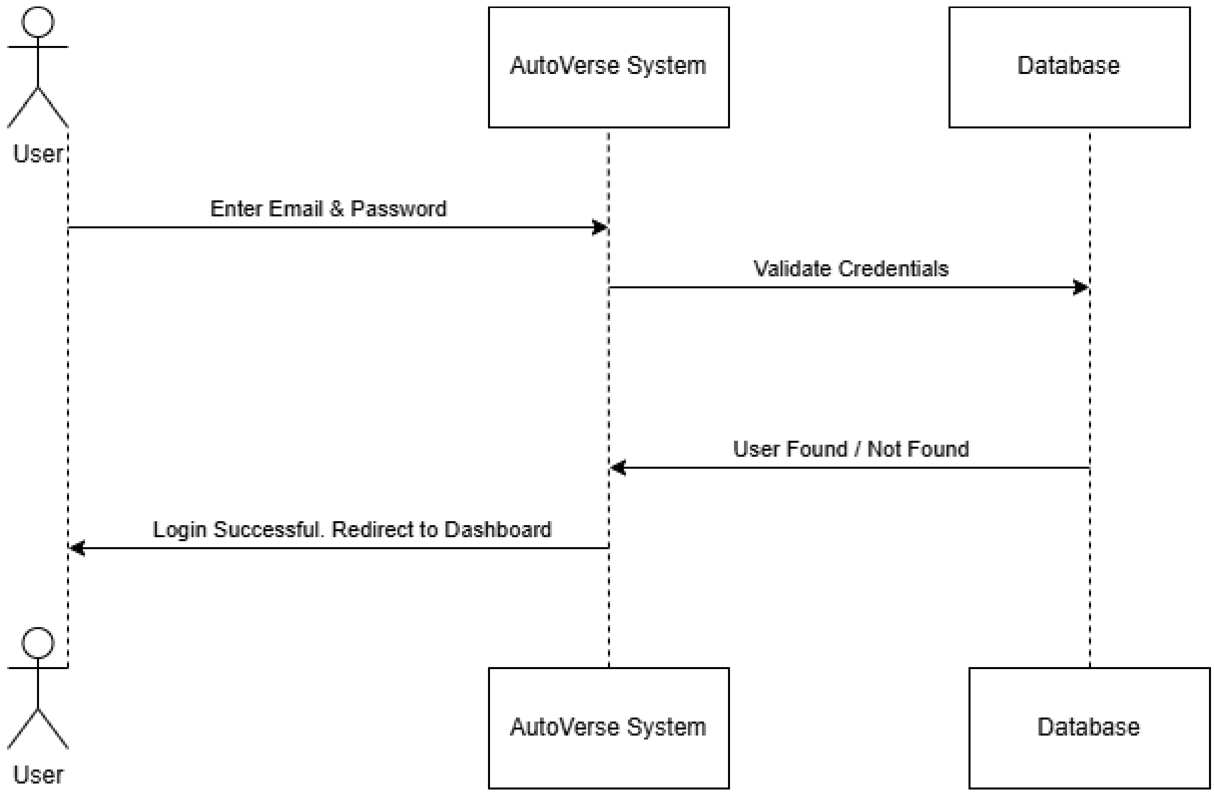


Figure 3.2 : User login

3.2.2 User Rent

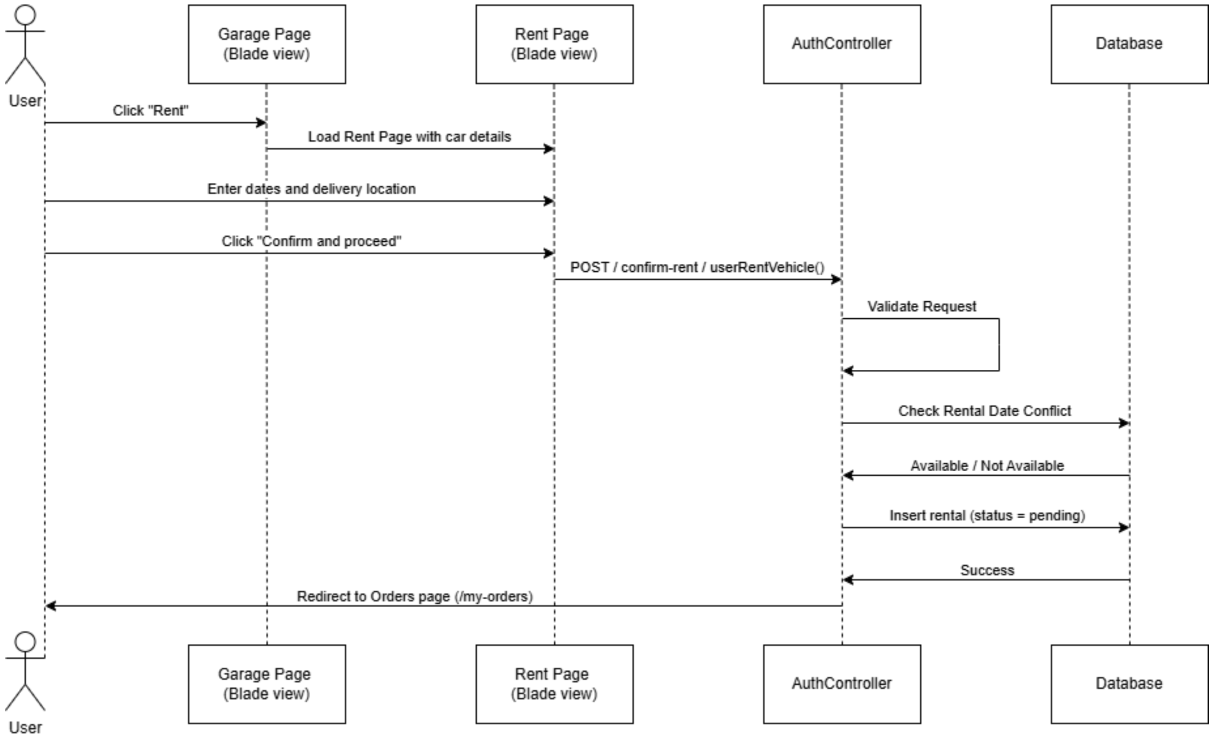


Figure 3.3 : User rent

3.2.3 Admin Adds A Car

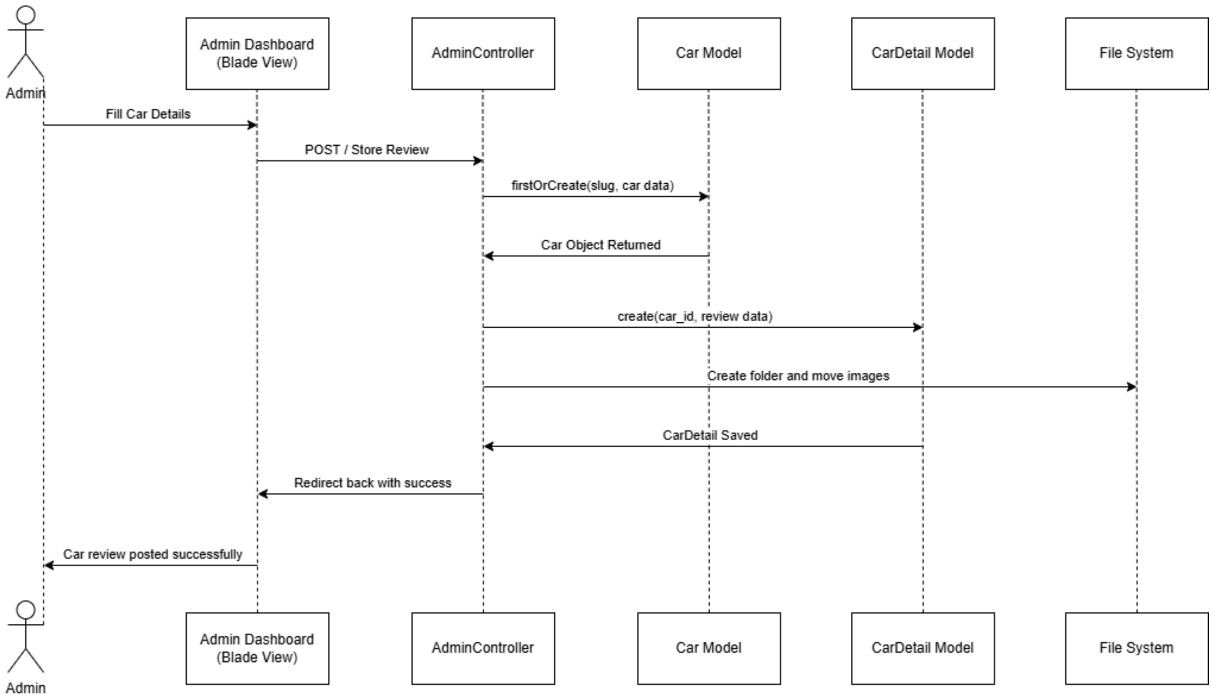


Figure 3.4 : Admin adds a car

3.2.4 Admin Approves Rental

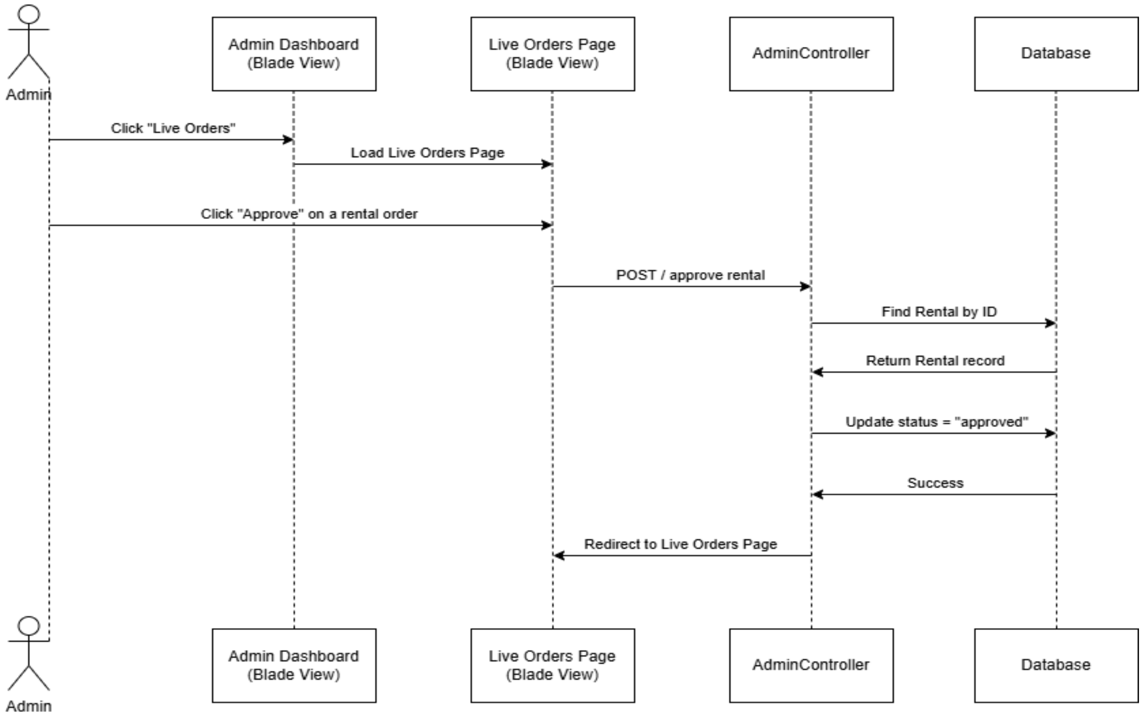


Figure 3.5 : Admin approves rental

3.2.5 User Adds Car To Wishlist

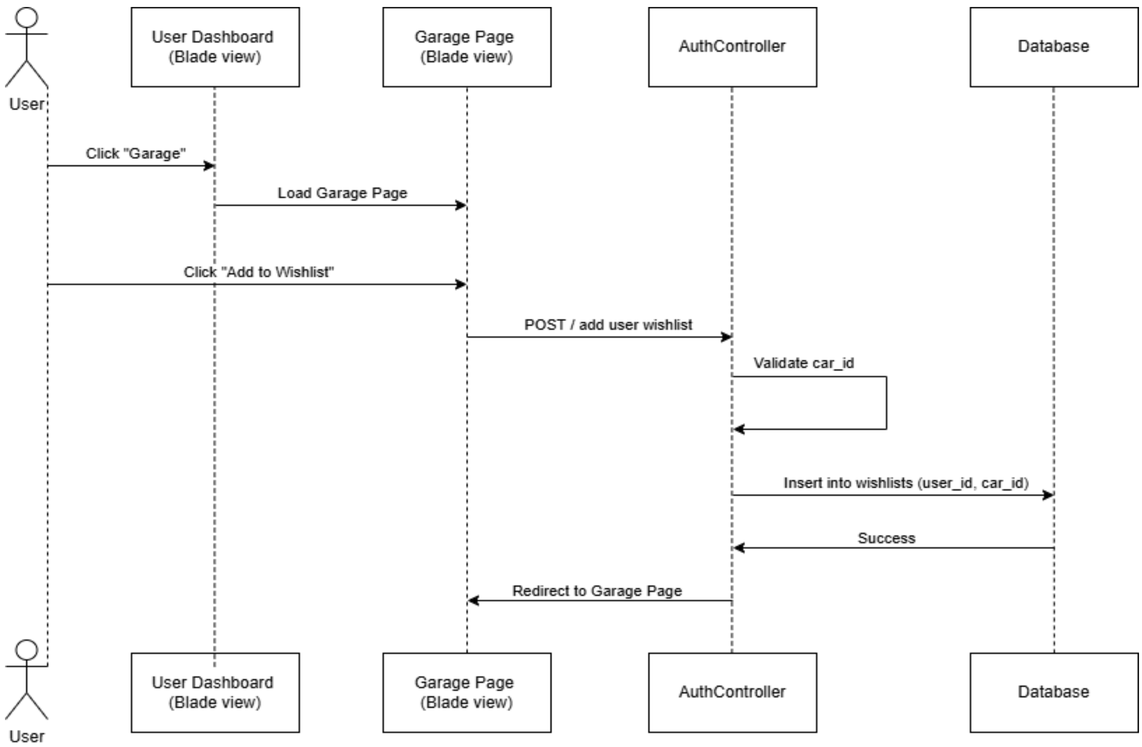


Figure 3.6 : User adds a car to wishlist

3.3 Entity Relationship Diagram

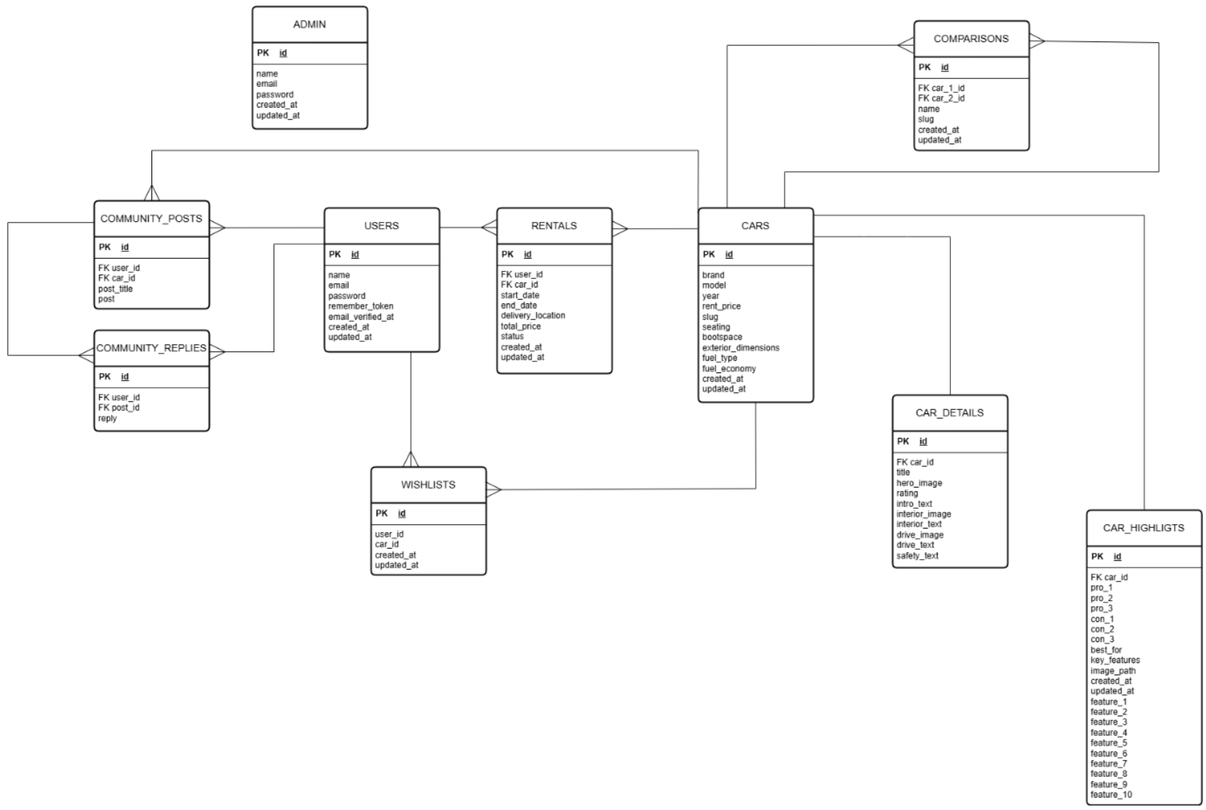


Figure 3.7 : Entity relationship diagram

3.4 Class Diagram

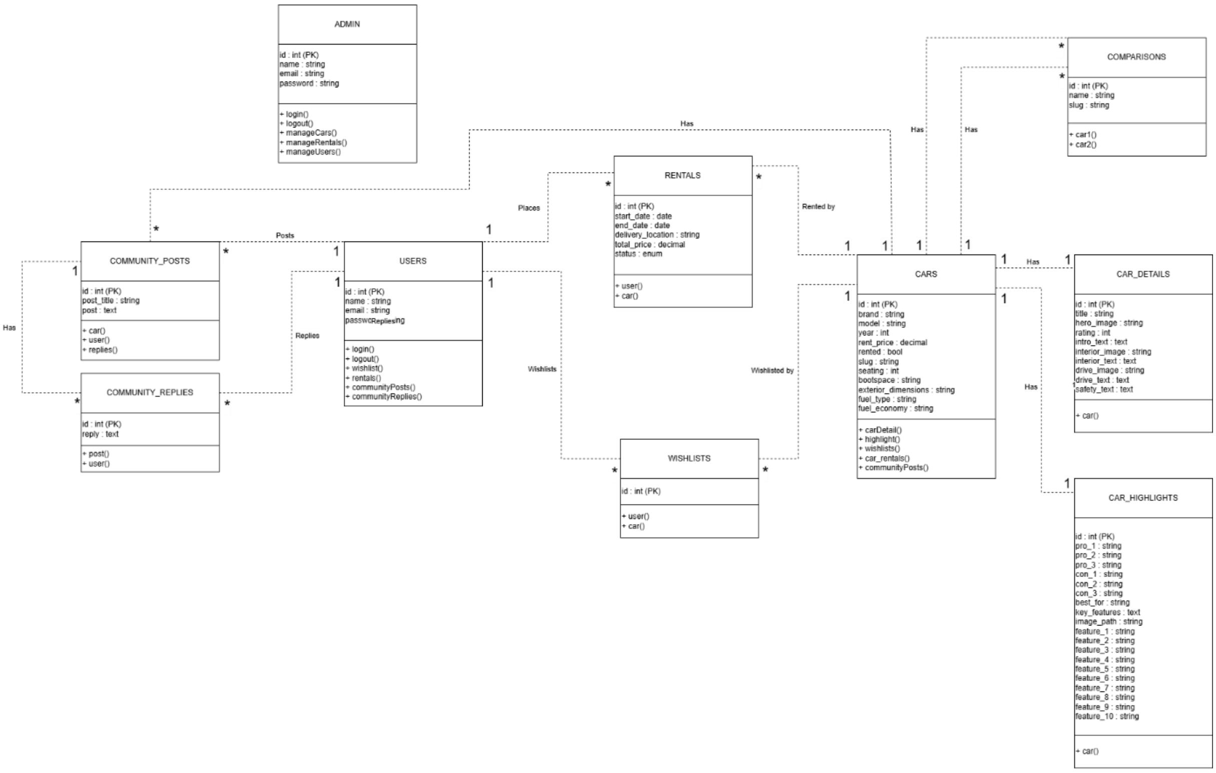


Figure 3.8 : Class diagram

3.5 Project Design

3.5.1 Input design

Inaccurate input data is one of the most common causes of errors in system processing. Error-free input can be achieved through effective input design. Input design is the process of converting user-provided data into a structured computer-based format. The primary objective of input design in **AutoVerse** is to make data entry simple, logical, secure, and error-free.

A well-designed input system should be user-friendly, interactive, responsive and tailored to meet user requirements. In **AutoVerse**, input forms are designed to ensure smooth interaction for both administrators and users. Proper validation techniques are implemented to maintain data accuracy and system reliability. The goal of input design is to reduce errors, improve usability, and ensure consistent data processing.

To achieve this, the following steps are considered:

- Allocating sufficient space for each input field

- Maintaining logical field sequence for better usability
- Defining clear formats for data entry (dates, numbers, email, etc.)
- Implementing both client-side and server-side validation
- Displaying proper error messages when invalid data is entered

When designing the input system, the following factors are considered:

- What data needs to be collected
- The medium of input (web-based forms)
- User guidance through labels, placeholders, and validation messages
- Methods for performing input validation and handling errors

Input design is a crucial part of the overall system architecture. Since incorrect input can lead to incorrect processing and output, strong validation and structured form design are implemented to ensure system accuracy, reliability, and efficiency.

1. User Registration Form

The user registration form collects necessary information to create a new user account.
It includes fields such as:

• Name
• Email Address
• Password
• Password Confirmation

Validation ensures that email addresses are unique and passwords meet security requirements.

2. User Login Form

The user login form allows registered users and administrators to access the system.
It includes:

• Email
• Password

Authentication mechanisms verify credentials before granting access to protected areas of the system.

3. Car Review Registration Form (Admin)

This form is used by the administrator to add new car reviews to the system.
It includes fields such as:

• Brand
• Model

- Year
- Rental Price
- Rating
- Seating Capacity
- Fuel Type
- Fuel Economy
- Boot Space
- Exterior Dimensions
- Review Content Sections
- Image Uploads

This form ensures structured storage of vehicle details and editorial content.

4. Car Highlights Form (Admin)

This form allows the administrator to add highlight details for a specific car.
It includes:

- Car Selection
- Pros
- Cons
- Best Suited For
- Key Features
- Highlight Image

This improves user understanding and comparison capabilities.

5. Car Comparison Form (Admin, User)

This form allows either the administrator or user to compare two vehicles.
It includes:

- Selection of Car 1
- Selection of Car 2

The system then generates a structured comparison view.

6. Edit Car Details Form (Admin)

The Edit Car Details form allows the administrator to modify existing vehicle information stored in the system. This form ensures that outdated or incorrect data can be updated efficiently.

7. Delete Car Form (Admin)

The Delete Car form allows the administrator to remove a car record from the system when necessary.

This action involves:

- Selecting the car to be deleted
- Confirming the deletion action

The system ensures that only authorized administrators can perform deletion operations. Once confirmed, the car record and its related details are removed from the database using controlled database operations.

8. Rental Request Form

This form allows users to request a vehicle rental. It includes:

- Start Date
- End Date
- Delivery Location

Date validation ensures logical rental periods, and the total rental cost is calculated accordingly.

9. Wishlist Functionality

Users can add or remove cars from their wishlist.
This action captures:

- User ID
- Car ID

This helps users track preferred vehicles.

10. Brand Filter Forms

Users can filter vehicles based on brand.
These forms improve usability and help users quickly find relevant cars.

3.5.2 Output Design

Output design plays a vital role in the **AutoVerse**, as it serves as the primary medium for communicating processed information to users and administrators. An effective output design strengthens user interaction with the system and supports informed decision-making related to vehicle selection, comparison, and rentals.

The outputs of the system include dynamic web pages, comparison views, dashboards, rental summaries, and confirmation messages. These outputs provide users with accurate, structured, and meaningful information while maintaining clarity and usability.

The output design of **AutoVerse** aims to provide users with timely, relevant, and well-organized information to enhance user experience and ensure efficient vehicle management.

The major system outputs include:

1. Car Review Display

This output presents detailed car reviews including:

• Brand and Model
• Rental Price
• Rating
• Technical Specifications
• Pros and Cons
• Highlight Features
• Editorial Review Sections
• Images

This allows users to view comprehensive information before making rental or comparison decisions.

2. Car Comparison Report

The comparison output displays a side-by-side analysis of two selected vehicles. It includes:

• Specifications comparison
• Pricing comparison
• Ratings comparison
• Feature differences

This output helps users evaluate vehicles effectively.

3. User Dashboard

The User Dashboard displays personalized information such as:

- Active Rentals
- Completed Rentals
- Cancelled Rentals
- Wishlist Vehicles
- Profile Information
- Access to communities

This output enables users to track their rental activities and preferences.

4. Rental Summary and Confirmation

When a rental request is submitted, the system generates a rental summary that includes:

- Rental Duration
- Delivery Location
- Total Rental Cost
- Rental Status

This output ensures transparency and record-keeping for users.

5. Admin Dashboard

The Admin Dashboard provides management-related outputs such as:

- List of All Cars
- Rental Requests
- Rental Status Updates
- Car Highlights
- Comparison Entries

This output allows administrators to monitor and manage system operations efficiently.

3.6 Table Design

The database design of **AutoVerse** is structured to efficiently manage vehicle information, users, rentals, and related operations. The tables are organized based on entities such as Users, Cars, Rentals, Highlights, Comparisons, and Wishlists.

The database follows normalization principles to minimize redundancy, maintain data integrity, and improve scalability.

The major tables in the system include:

1. Users Table

The Users table stores details of all registered users. It includes:

- User ID
- Name
- Email
- Password

Each user can create multiple rentals and wishlist entries.

2. Admin table

The admin table stores details of the administrator. It includes:

- Admin ID
- Name
- Email
- Password

3. Cars Table

The Cars table stores core vehicle information. It includes:

- Car ID
- Brand
- Model
- Year
- Rental Price
- Availability Status
- Slug
- Seating
- Bootspace
- Exterior Dimensions
- Fuel types
- Fuel economy

Each car can have associated details, highlights, comparisons, and rentals.

4. Car Details Table

The Car Details table stores extended editorial and specification information for each vehicle. It includes:

- Car ID (Foreign Key)
- Review Content Sections
- Ratings
- Technical Specifications
- Images

This table maintains a one-to-one relationship with the Cars table.

5. Rentals Table

The Rentals table stores all rental transactions. It includes:

- Rental ID
- User ID (Foreign key)

- Car ID (Foreign key)
- Start Date
- End Date
- Delivery Location
- Total Price
- Rental Status

Each rental is associated with one user and one car.

6. Wishlists Table

The Wishlists table stores vehicles saved by users. It includes:

- Wishlist ID
- User ID (Foreign key)
- Car ID (Foreign key)

This table represents a many-to-many relationship between users and cars.

7. Highlights Table

The Highlights table stores highlight information related to specific cars. It includes:

- Highlight ID
- Car ID
- Pros
- Cons
- Key Features
- Highlight Image

8. Comparisons Table

The Comparisons table stores predefined comparison records. It includes:

- Comparison ID
- Car 1 ID
- Car 2 ID
- Comparison Title

9. Community Posts table

The Community Posts table stores user posts for a particular vehicle

- Community Post ID
- User ID

- Car ID
- Post title
- Post

10. Community Replies table

The community Replies table stores replies to user posts

- Community Replies ID
- Post ID
- User ID
- Reply

The database structure ensures minimal redundancy and maintains referential integrity using primary and foreign key constraints. By following normalization principles, the system ensures efficient data storage, reduced anomalies, and improved performance as the number of users and vehicles increases. This structured database design enables **AutoVerse** to manage vehicle reviews, rental operations, and user interactions effectively.

4. TESTING

Software testing is a crucial phase in the development of **AutoVerse**. It ensures that the application functions correctly, meets user requirements, and operates without unexpected errors. The testing process in **AutoVerse** was primarily conducted through manual testing and iterative debugging during development. Each module was individually implemented, executed, verified, and refined until it functioned correctly. This approach ensured early detection of errors and improved overall system quality.

4.1 Unit Testing

Unit testing involves testing individual components of the system in isolation to verify their correctness.

In **AutoVerse**, individual modules such as:

- User Registration
- User Login
- Car Creation Module
- Rental Processing Module
- Wishlist Functionality
- Comparison Module

were tested independently during development.

Each feature was implemented and immediately executed in the browser to verify expected behavior. Errors such as validation failures, routing issues, database relationship problems, and logic errors were identified and corrected before proceeding to the next module.

Purpose:
To ensure that each unit of the system performs its intended function correctly.

4.2 Integration Testing

Integration testing ensures that different modules work together properly.

In **AutoVerse**, integration testing was performed by verifying:

- User authentication working with dashboard and community access
- Rental module correctly storing data in the database
- Wishlist functionality correctly linking users and cars
- Admin operations reflecting properly on the user interface
- Relationships between Cars, Car Details, Rentals, and Users

This testing ensured smooth interaction between frontend views, backend logic, and database operations.

Purpose:
To detect issues arising from interaction between different modules.

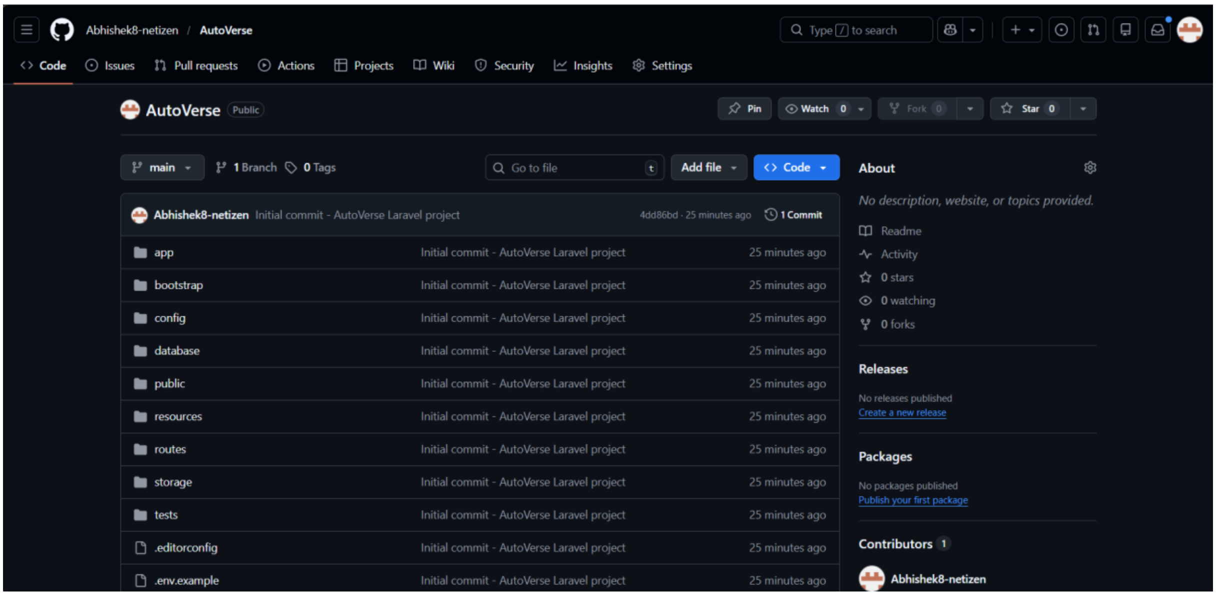
5. DEPLOYMENT

5.1 Folder Structure

```
AutoVerse/
│
├── app/
│   │
│   ├── Models/
│   │   ├── Admin.php
│   │   ├── Car.php
│   │   ├── CarDetail.php
│   │   ├── CarHighlight.php
│   │   ├── CommunityPosts.php
│   │   ├── CommunityReply.php
│   │   ├── Comparison.php
│   │   ├── Rental.php
│   │   ├── User.php
│   │   └── Wishlist.php
│   │
│   ├── Http/
│   │   │
│   │   ├── Controllers/
│   │   │   ├── AdminController.php
│   │   │   ├── AuthController.php
│   │   │   ├── CommunityController.php
│   │   │   ├── ComparisonController.php
│   │   │   ├── Controller.php
│   │   │   ├── PageController.php
│   │   │   ├── ProfileController.php
│   │   │   ├── RentalController.php
│   │   │   └── ReviewController.php
│   │   │
│   │   └── Middleware/
│   │       ├── AdminAuth.php
│   │       └── CompletedRentalOnly.php
│
├── database/
│   ├── migrations/
│   └── seeders/
│
├── resources/
│   ├── views/
│   ├── css/
│   └── js/
│
├── public/
│   ├── images/
│   ├── css/
│   └── js/
│
├── routes/
│   └── web.php
│
├── storage/
│
├── config/
│
├── bootstrap/
│
├── .env
├── composer.json
└── artisan
```

Figure 5.1 : Folder structure

6. GIT HISTORY



7. CONCLUSION

The **AutoVerse** system is an efficient solution for managing car rentals, reviews, and user interactions in a streamlined and automated way. By centralizing car data, user rentals, wishlists, and editorial reviews, the system improves operational efficiency. Users can easily browse, compare, and rent cars, while the system ensures accurate tracking of rentals, and availability.

Thoroughly tested, **AutoVerse** is reliable and user-friendly, requiring minimal technical expertise for both admins and users. Features like wishlist management, car comparisons, and rental lifecycle tracking enhance the overall user experience.

Overall, **AutoVerse** enhances the car rental process by providing a modern, organized, and interactive platform that benefits both users and administrators. The system ensures timely availability of cars, efficient handling of rentals, and a robust framework for future enhancements like community engagement, analytics, payment gateways and enhanced filters.

7.1 Future Enhancements

- Advanced Analytics: Add dashboards to track popular cars, rental trends, user behavior, and wishlist activity. This would help admins make data-driven decisions.
- Mobile App / Improved Responsiveness: Enhance the system with a dedicated mobile app or further refine the website layout for small screens, making browsing, comparing, and renting cars smoother on all devices.
- Payment & Invoice Automation: Integrate payment gateways and automatically generate invoices for rentals to streamline the rental process and improve user experience.
- Enhanced Search & Filtering: Improve the car search system with advanced filters (price range, fuel type, rating, availability dates) to make it easier for users to find their preferred car quickly.

8. APPENDIX

8.1 Minimum Software Requirements

The software requirements for **AutoVerse** are listed below:

- **Software Development Platform :** PHP 8.2, Laravel 12, HTML5, CSS3, JavaScript
- **Operating System :** Microsoft Windows 10 or later, macOS, or Linux (any modern distro)
- **Database :** PostgreSQL (via pgAdmin 4)
- **Web Server :** Laravel's built-in server (php artisan serve) or any compatible web server like Apache/Nginx
- **PHP Version :** 8.2 or later
- **HTML Version :** 5.0
- **Frontend Framework :** Bootstrap 5

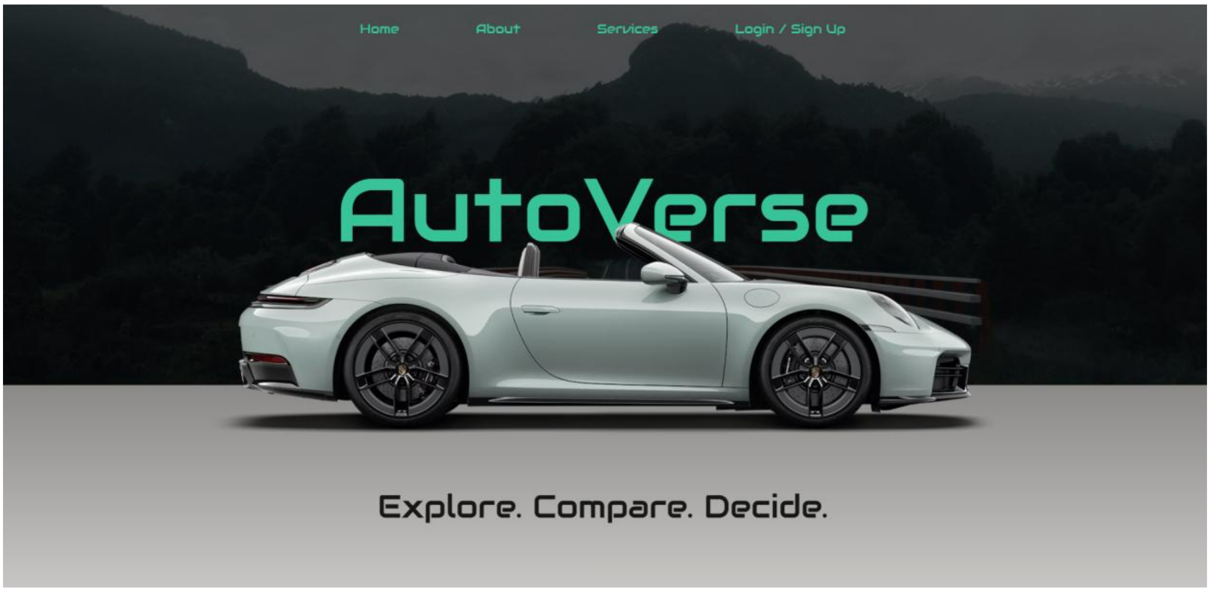
8.2 Minimum Hardware Requirements

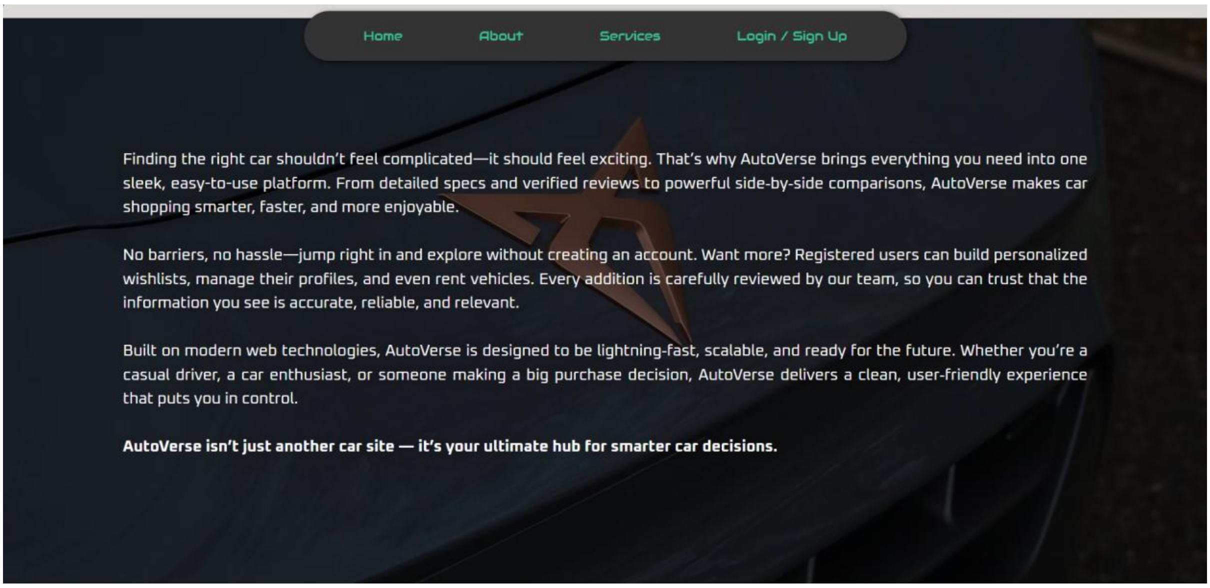
The initial deployment is expected to support minimum configuration as illustrated

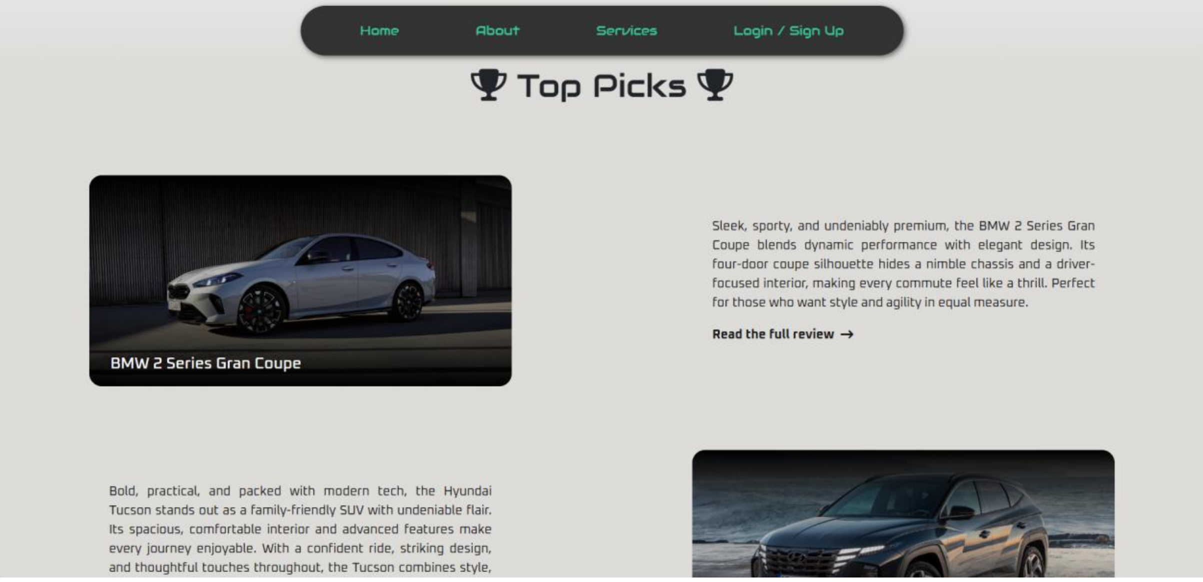
- **Processor** **:** Intel i5 / AMD Ryzen 5 or higher
- **Memory (Storage)** **:** 256 GB SSD or above
- **RAM** **:** 8 GB or above
- **Network** **:** High-speed broadband connection
- **Display** **:** 1920 × 1080 resolution or higher

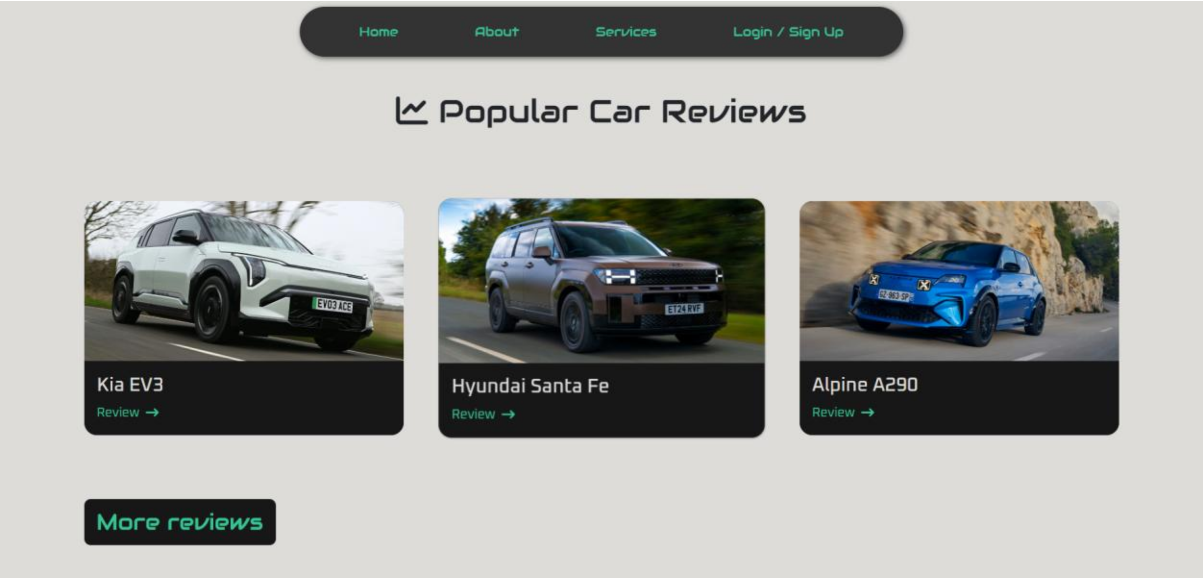
9. SCREENSHOTS

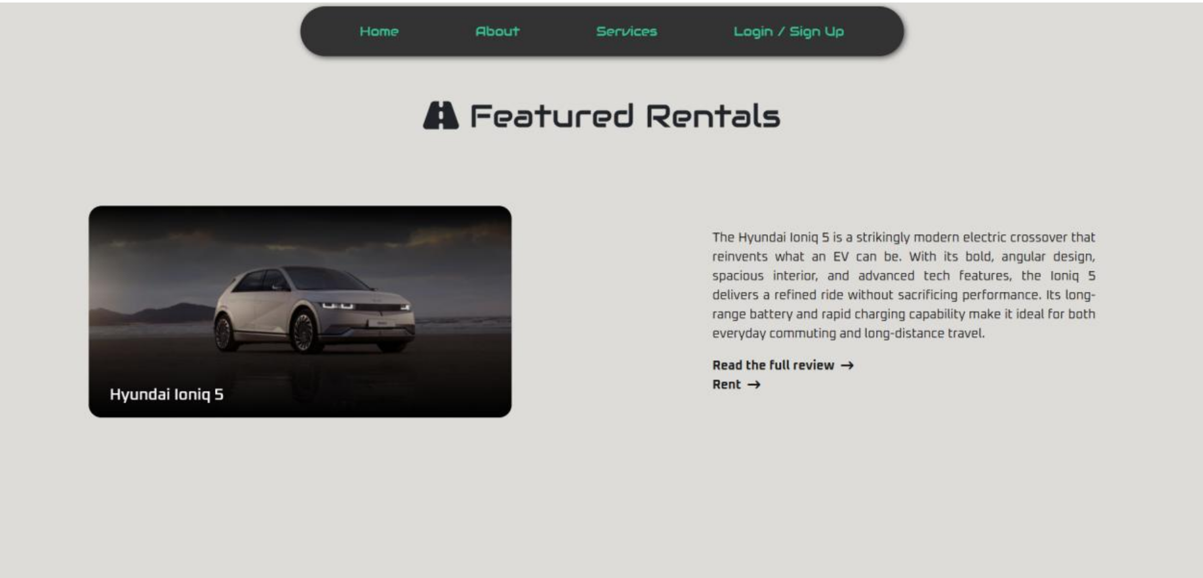
Landing Page



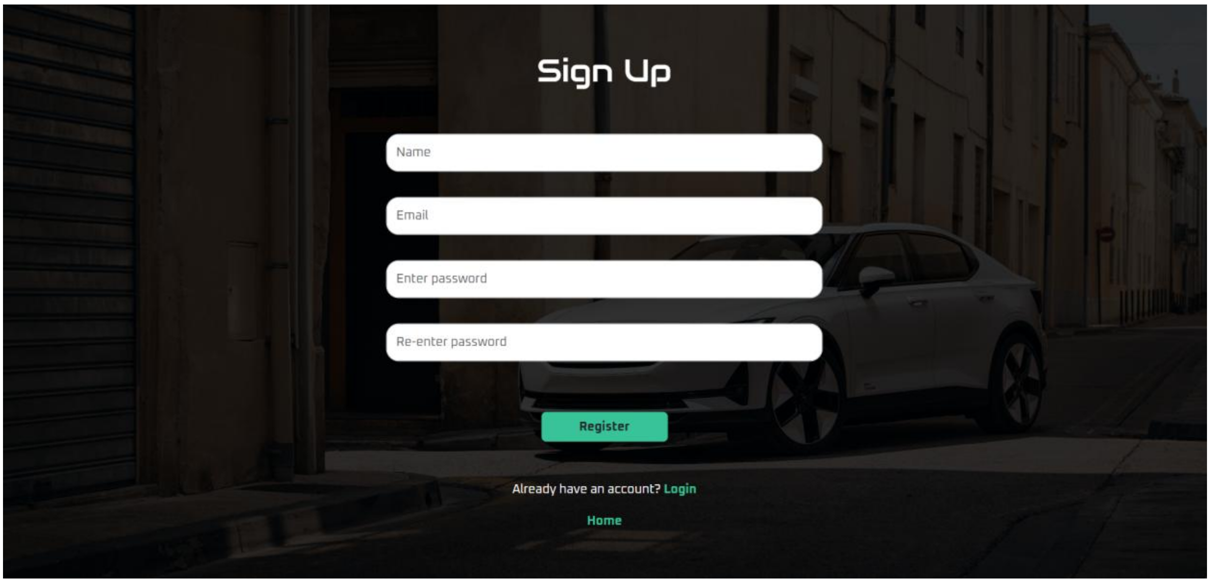



Home
About
Services
Login / Sign Up
Top Picks
BMW 2 Series Gran Coupe
Sleek, sporty, and undeniably premium, the BMW 2 Series Gran Coupe blends dynamic performance with elegant design. Its four-door coupe silhouette hides a nimble chassis and a driver-focused interior, making every commute feel like a thrill. Perfect for those who want style and agility in equal measure.
Read the full review →
Bold, practical, and packed with modern tech, the Hyundai Tucson stands out as a family-friendly SUV with undeniable flair. Its spacious, comfortable interior and advanced features make every journey enjoyable. With a confident ride, striking design, and thoughtful touches throughout, the Tucson combines style,

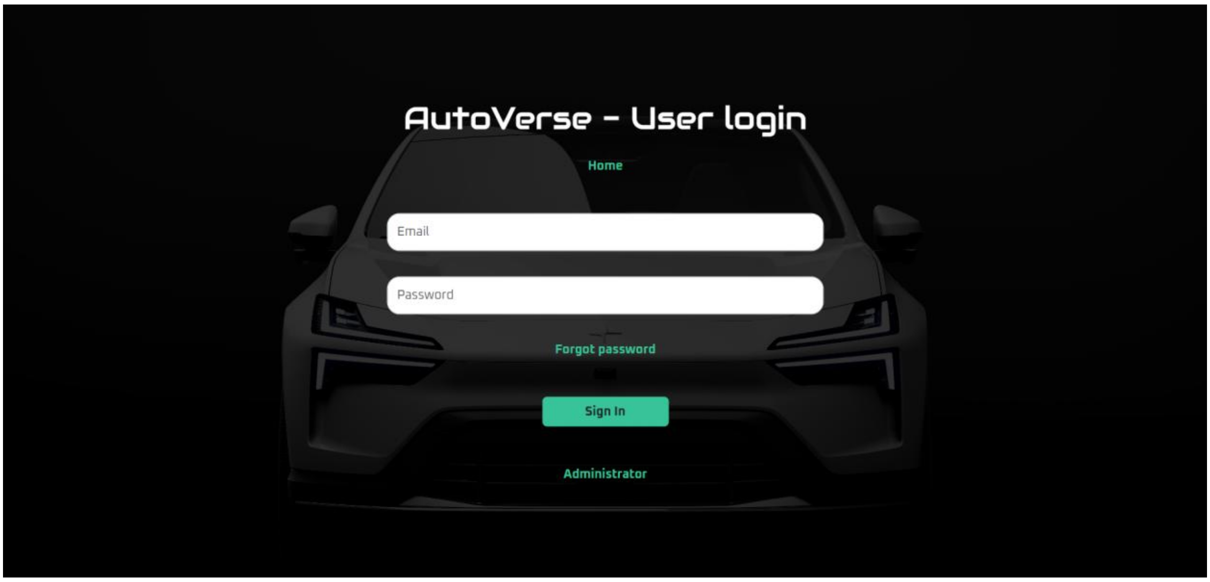

Home
About
Services
Login / Sign Up
Popular Car Reviews
Kia EV3
Review →
Hyundai Santa Fe
Review →
Alpine A290
Review →
More reviews


Home
About
Services
Login / Sign Up
Featured Rentals
Hyundai Ioniq 5
The Hyundai Ioniq 5 is a strikingly modern electric crossover that reinvents what an EV can be. With its bold, angular design, spacious interior, and advanced tech features, the Ioniq 5 delivers a refined ride without sacrificing performance. Its long-range battery and rapid charging capability make it ideal for both everyday commuting and long-distance travel.
Read the full review →
Rent →

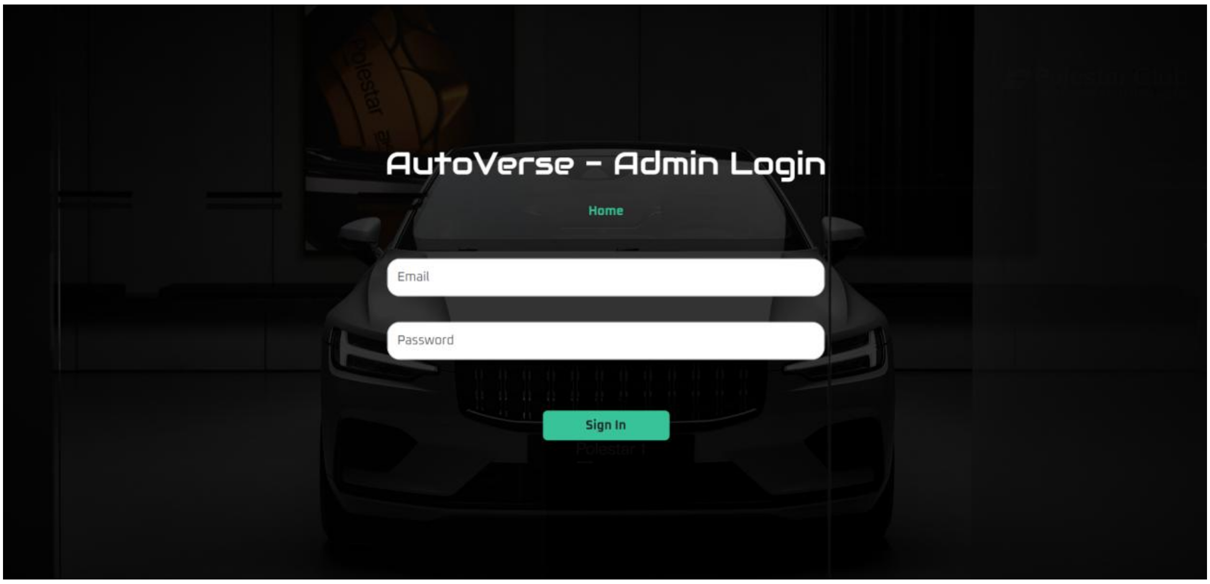
Registration Page



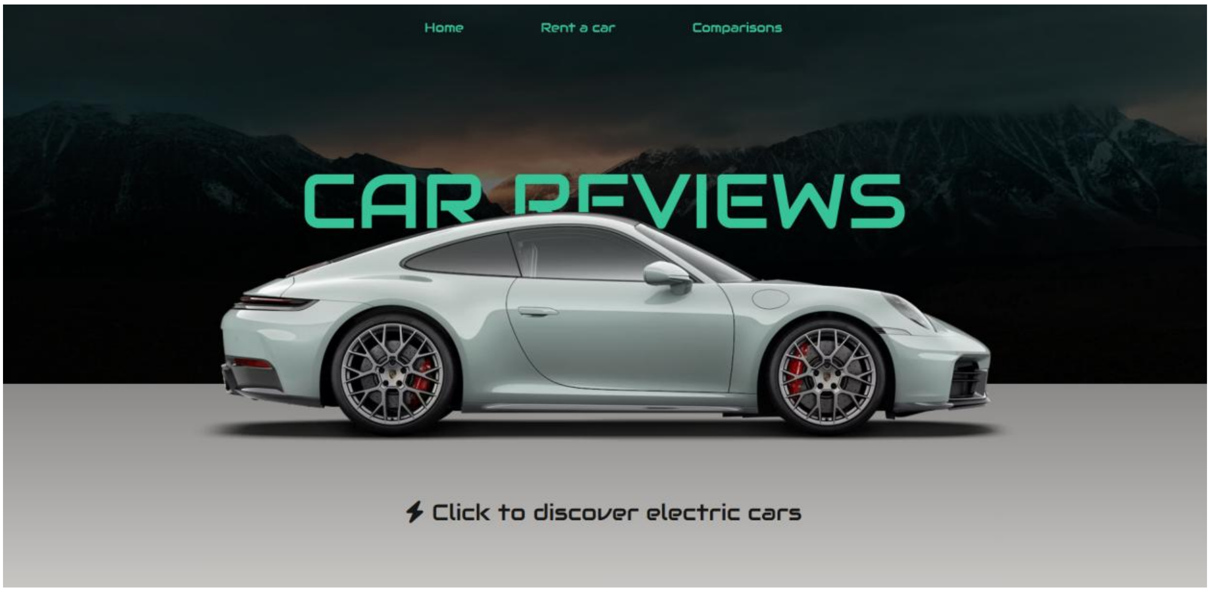
User Login Page

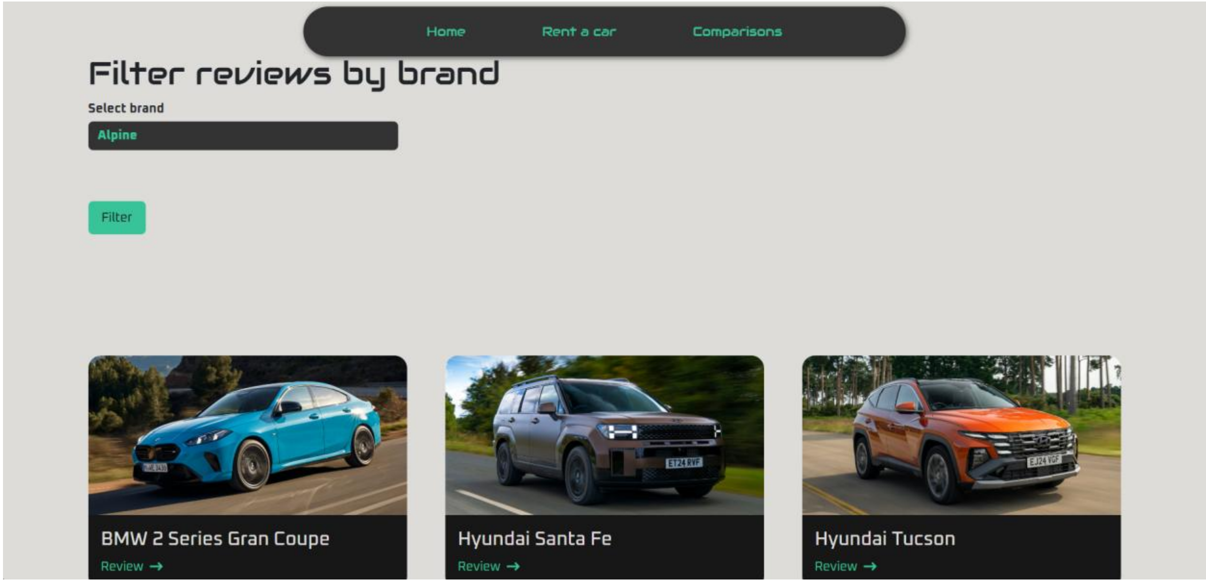


Admin Login Page

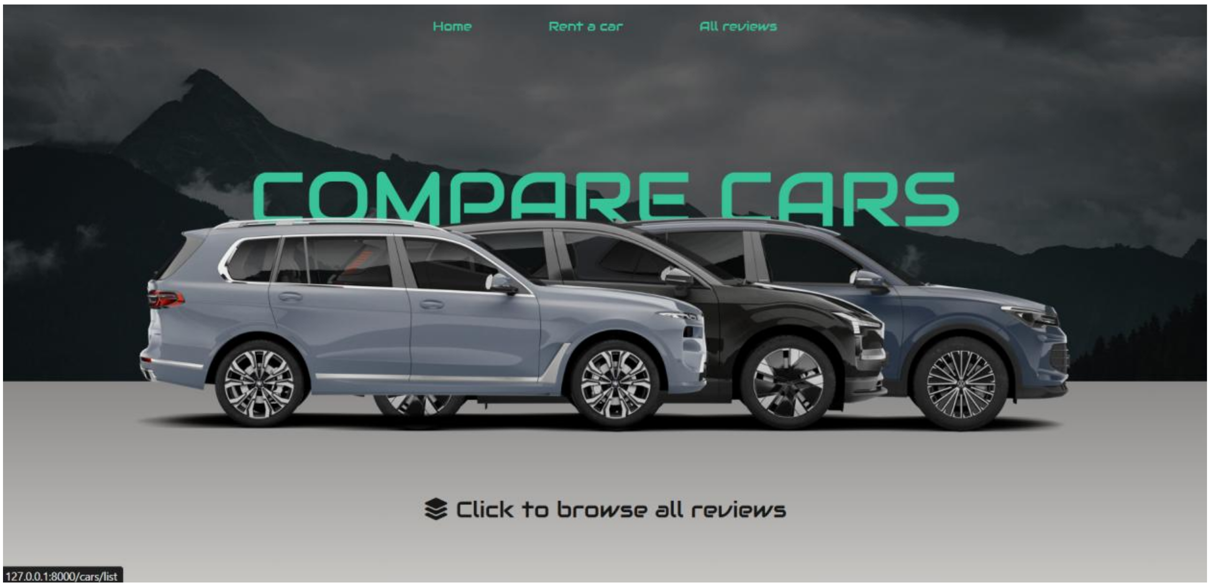


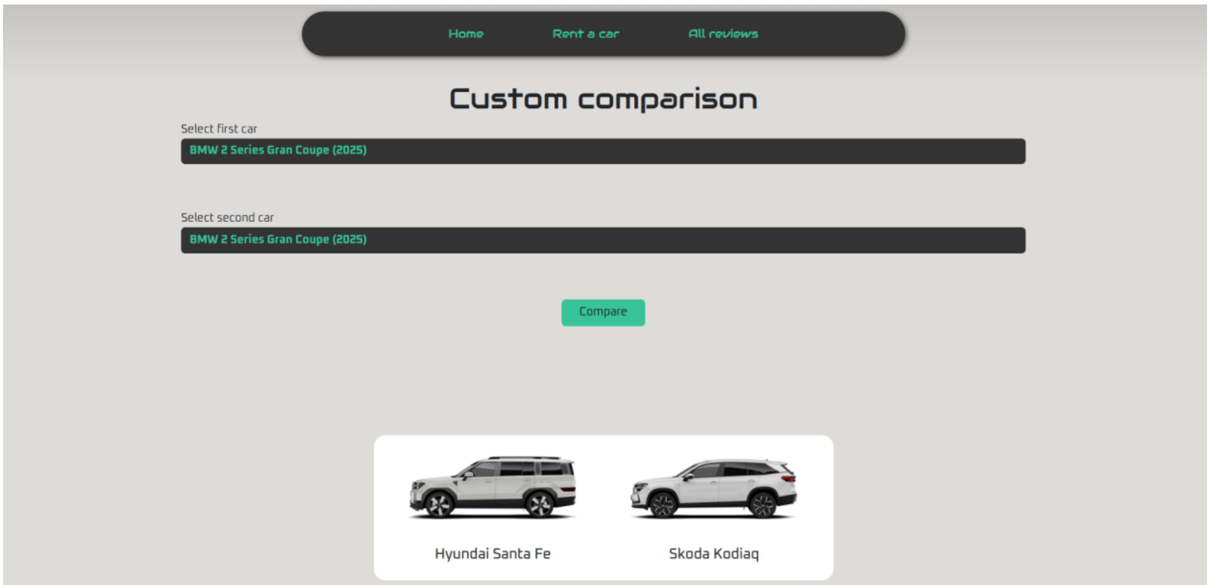
Reviews Page



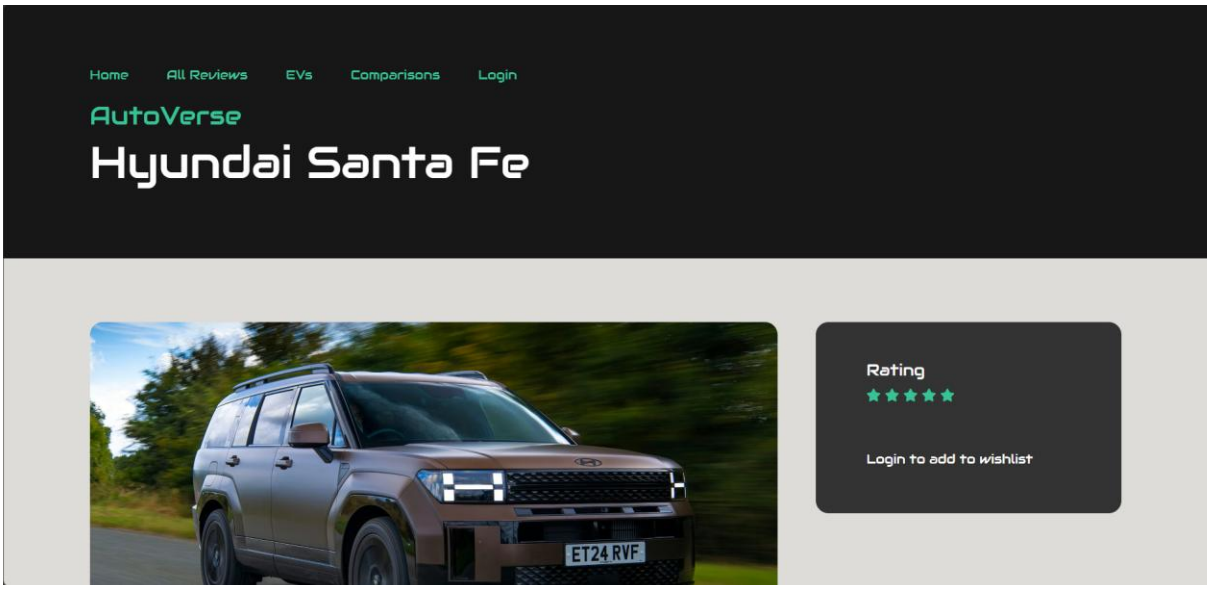


Comparisons Page





Individual Review Page



Show specifications

Number of seats : 7

Boot space, seats up : 621 - 628 litres

Exterior dimensions (L x W x H) : 4,830 mm x 1,900 mm x ?? mm

Available fuel types : Hybrid

Fuel economy : 38.2 - 166.0 mpg

Is the Hyundai Santa Fe a good car?

The Hyundai Santa Fe is one of the company's stalwarts - it's been around for nearly 25 years and during that time morphed from a slightly agricultural off-roader into a proper family car. The latest model, though, adds in a real dose of Star Wars styling and makes what would otherwise be a pretty dull-but-worthy SUV into something that's actually really cool. Alternatives include the closely-related Kia Sorento, as well as other three-row SUVs such as the Skoda Kodiaq and Nissan X-Trail. However, the new Santa Fe feels so upmarket that it's easy to imagine you might cross-shop it with something like a Land Rover Discovery or Volvo XC90 - it's that good.

The Santa Fe's boxy body is immediately eye-catching, and follows the design language of other great-looking Hyundais such as the Ioniq 5. It's futuristic but functional, with a square back that maximises the amount of interior space. All-round LED lights and an array of excellent colours - brown, orange or green are on the list as well as more ordinary shades of grey or silver - complete the forward-thinking approach

Interior style and infotainment

occasionally spoils the silence is the engine, which can be a bit raucous if you ask for a sudden burst of acceleration.

Power delivery isn't as effortless as it is in something like a Land Rover Discovery, simply because the Santa Fe doesn't have as much power on tap - but it's still more than comfortable to cruise all day at the legal limit and there's enough in reserve for a slip road or a quick overtake. Hyundai's assisted driving features also work quite well, with adaptive cruise control and lane-keeping aids that make for pretty smooth progress. We wouldn't bother using the automatic lane-changing, though - it's far too hesitant..

Safety and security

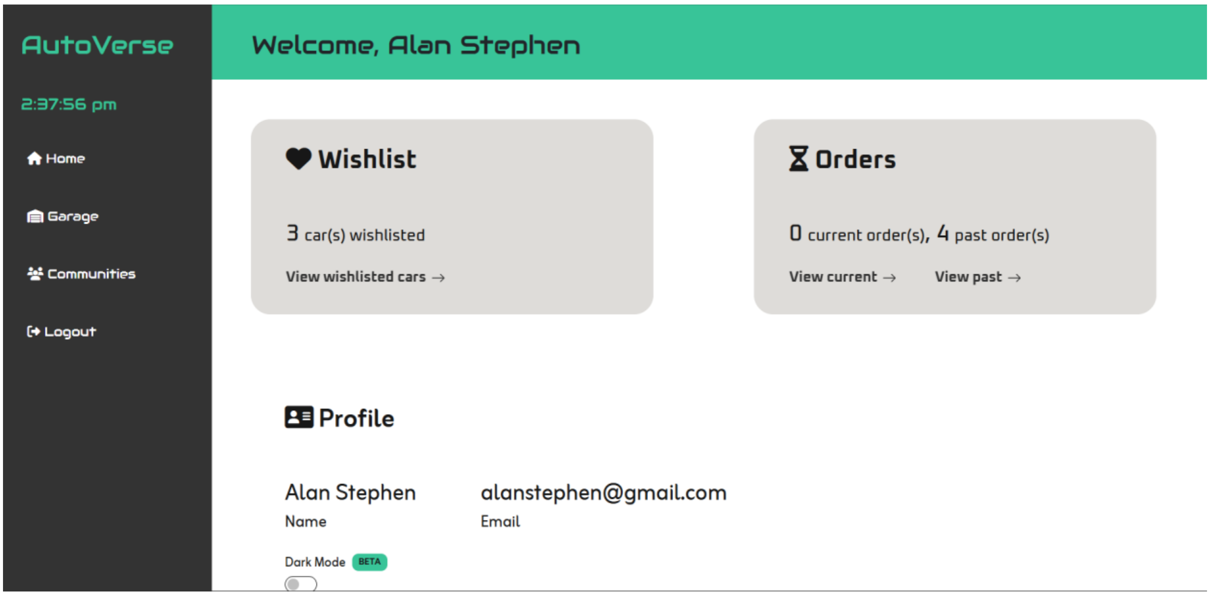
Euro NCAP crash-tested the Santa Fe, scoring it four stars overall and five stars with a 'safety pack'. This refers to the 'Smartsense' pack which is a reasonably-priced option on base models and standard on the upper trims. It adds more sophisticated adaptive cruise control, autonomous emergency braking, and blind-spot monitoring tech.

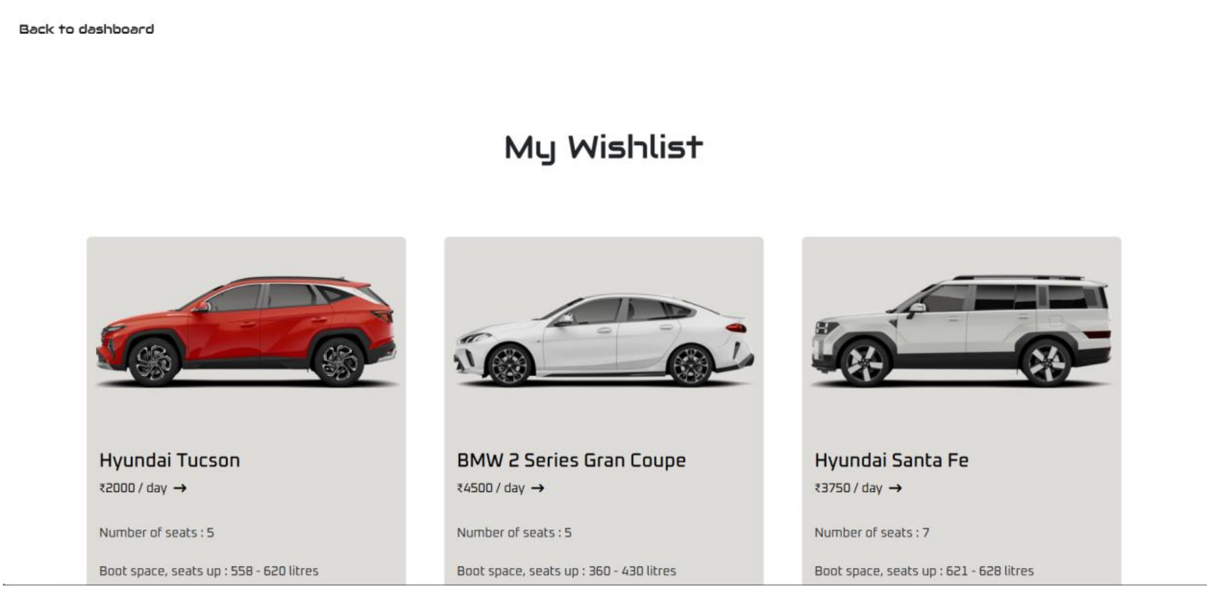
All Santa Fes come with a good basic level of safety aids, including driver attention monitoring, speed limit assist and lane-keeping aids. All cars also get blind-spot monitoring, full LED lights and a reversing camera. It's a shame it couldn't score the full five stars outright, though you should be reassured by its impressive 84% and 88% scores in adult and child occupant protection respectively.

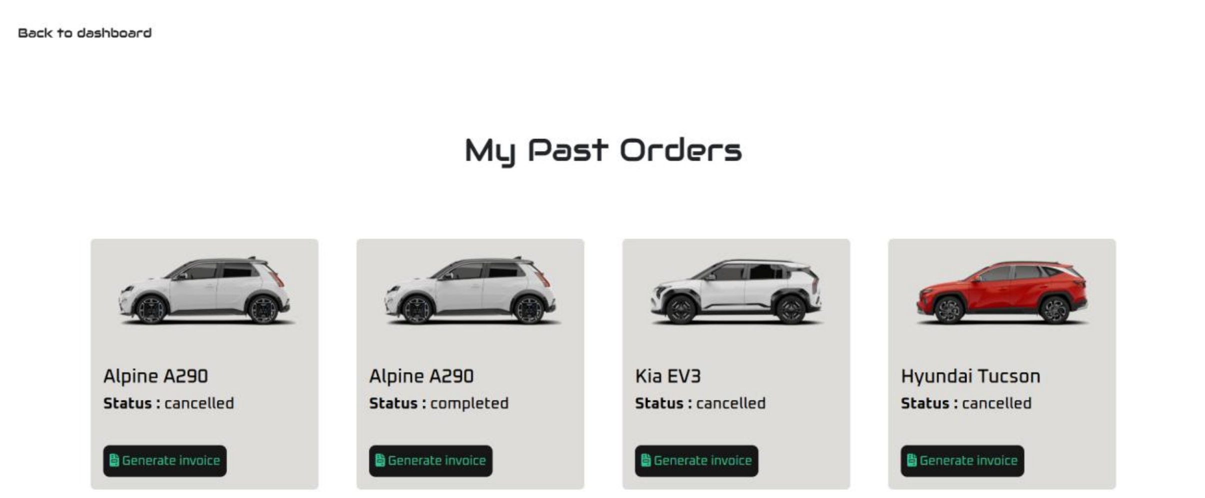
Back to all reviews   Compare cars   Explore EVs

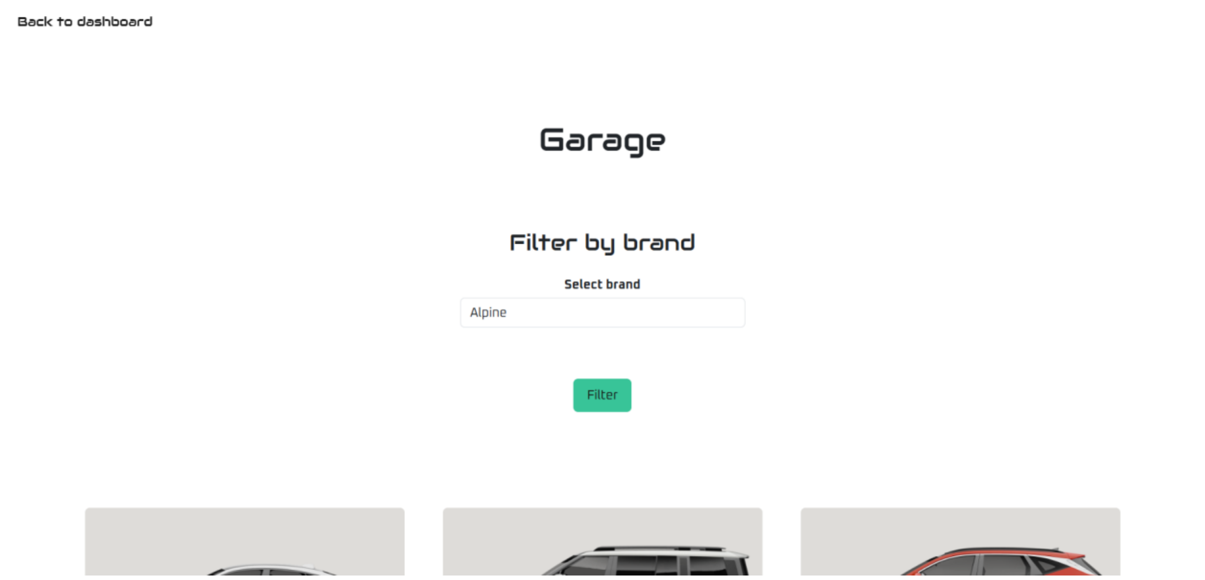
Individual Comparison Page

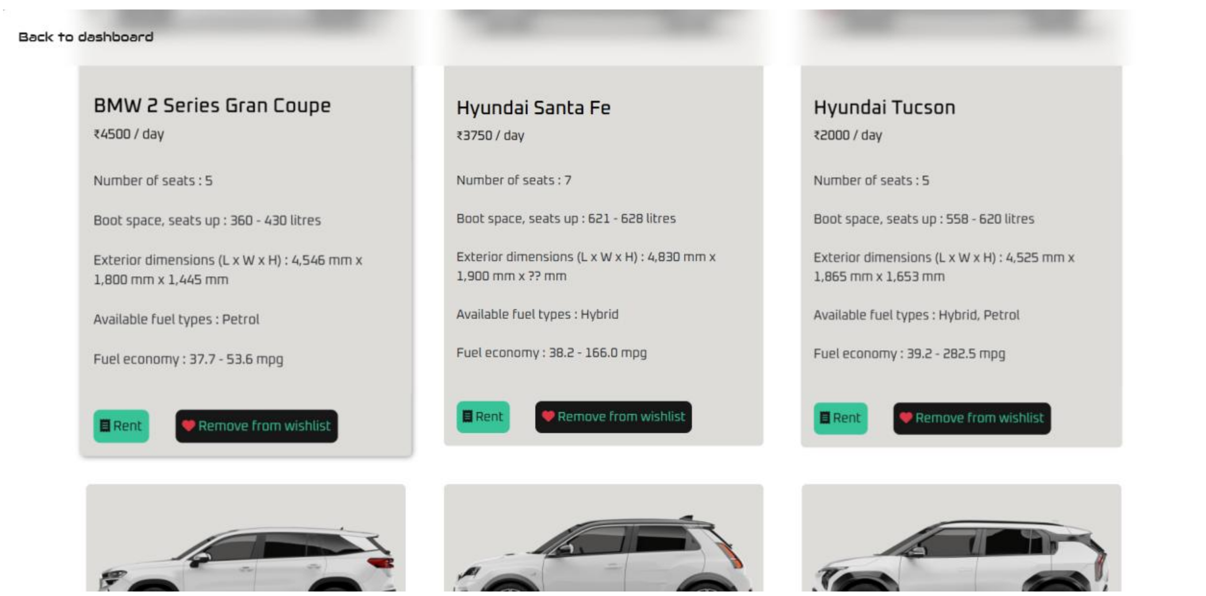
User Dashboard

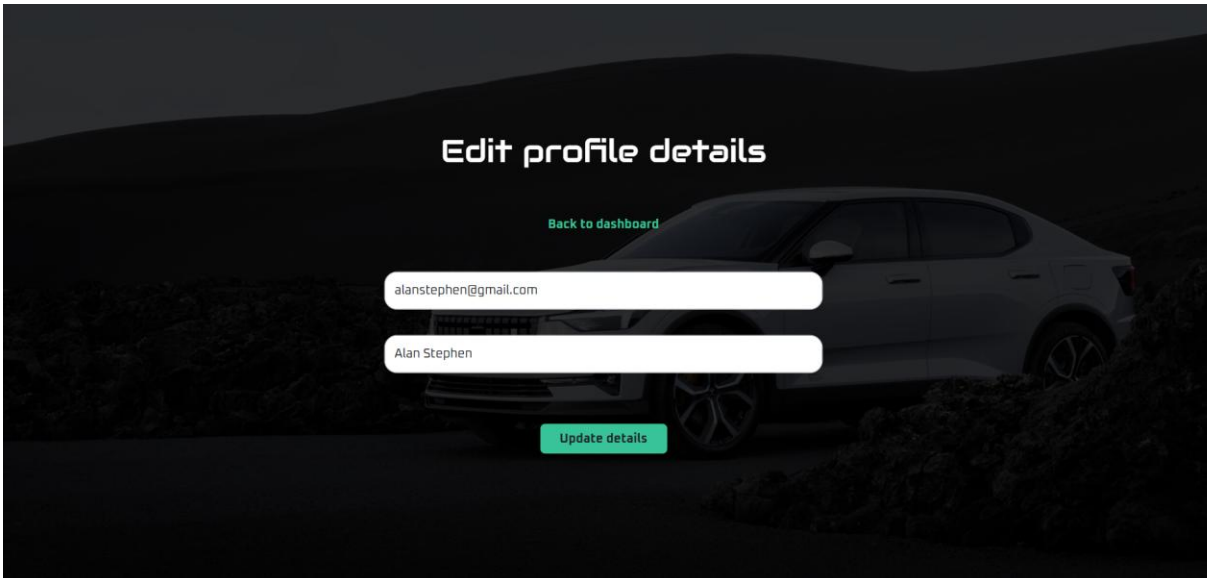


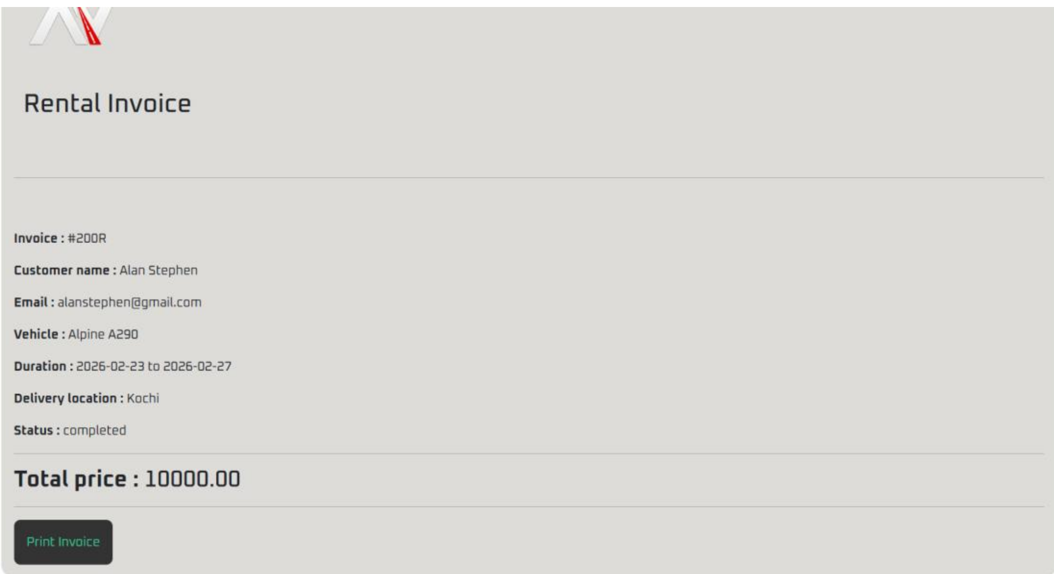



Back to dashboard
My Past Orders
Alpine A290
Status : cancelled
Generate invoice
Alpine A290
Status : completed
Generate invoice
Kia EV3
Status : cancelled
Generate invoice
Hyundai Tucson
Status : cancelled
Generate invoice

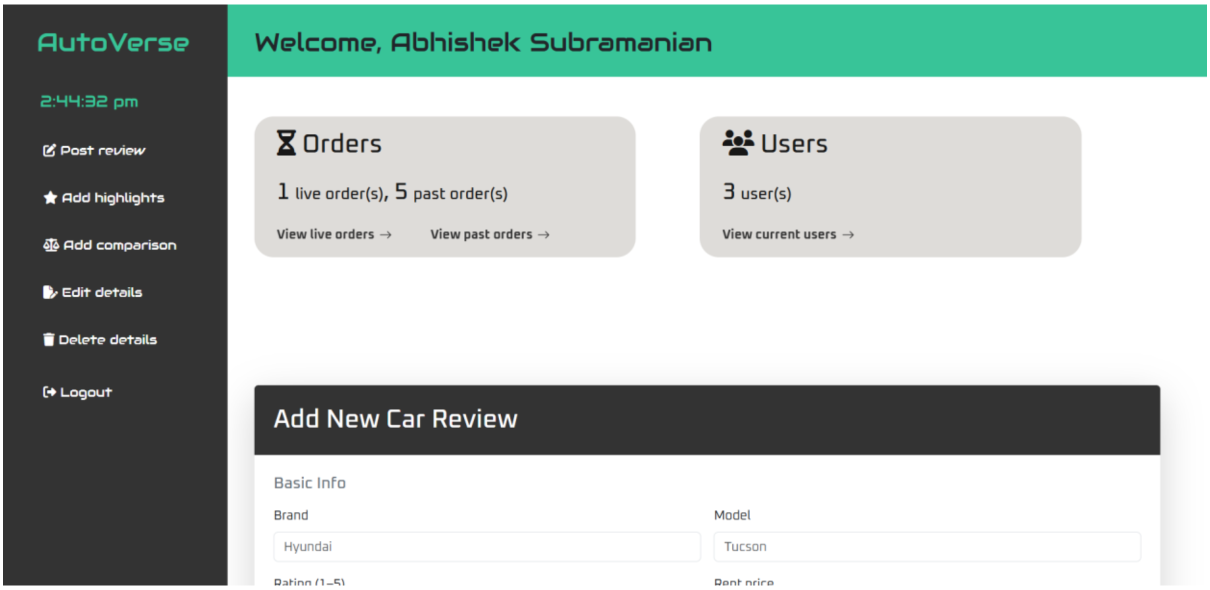

Back to dashboard
Garage
Filter by brand
Select brand
Alpine
Filter

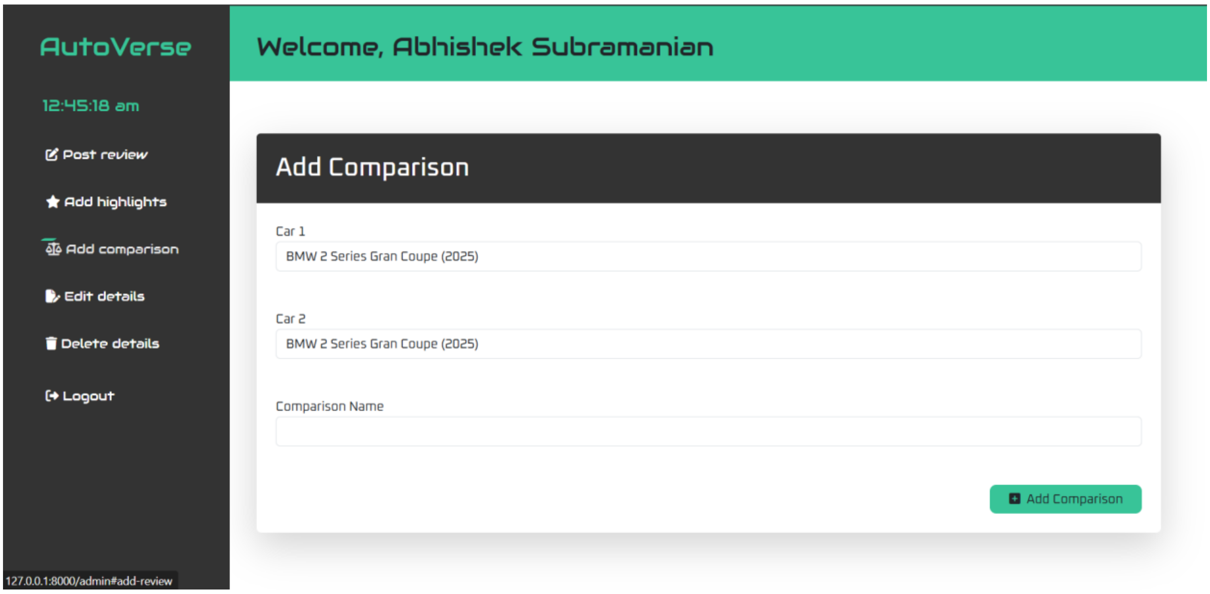

Back to dashboard
BMW 2 Series Gran Coupe
₹4500 / day
Number of seats : 5
Boot space, seats up : 360 - 430 litres
Exterior dimensions (L x W x H) : 4,546 mm x 1,800 mm x 1,445 mm
Available fuel types : Petrol
Fuel economy : 37.7 - 53.6 mpg
Rent
Remove from wishlist
Hyundai Santa Fe
₹3750 / day
Number of seats : 7
Boot space, seats up : 621 - 628 litres
Exterior dimensions (L x W x H) : 4,830 mm x 1,900 mm x ?? mm
Available fuel types : Hybrid
Fuel economy : 38.2 - 166.0 mpg
Rent
Remove from wishlist
Hyundai Tucson
₹2000 / day
Number of seats : 5
Boot space, seats up : 558 - 620 litres
Exterior dimensions (L x W x H) : 4,525 mm x 1,865 mm x 1,653 mm
Available fuel types : Hybrid, Petrol
Fuel economy : 39.2 - 282.5 mpg
Rent
Remove from wishlist

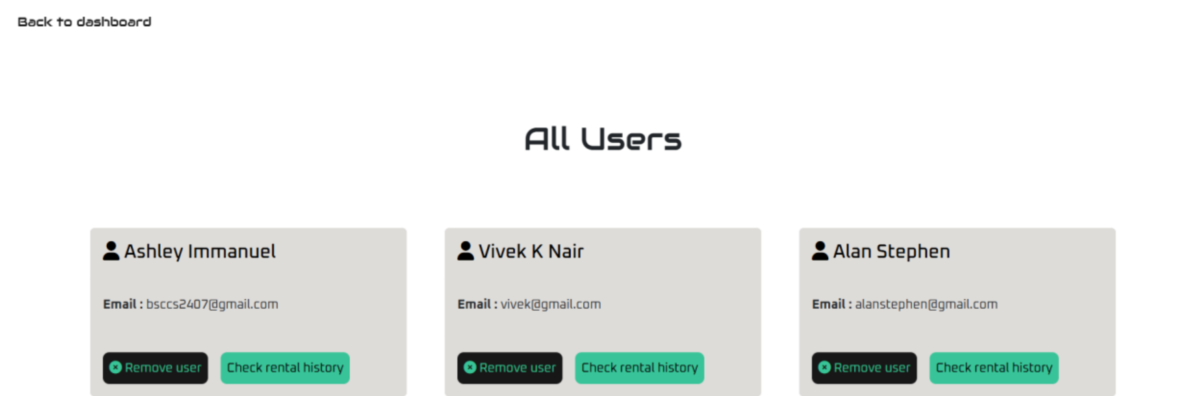

Edit profile details
Back to dashboard
alanstephen@gmail.com
Alan Stephen
Update details

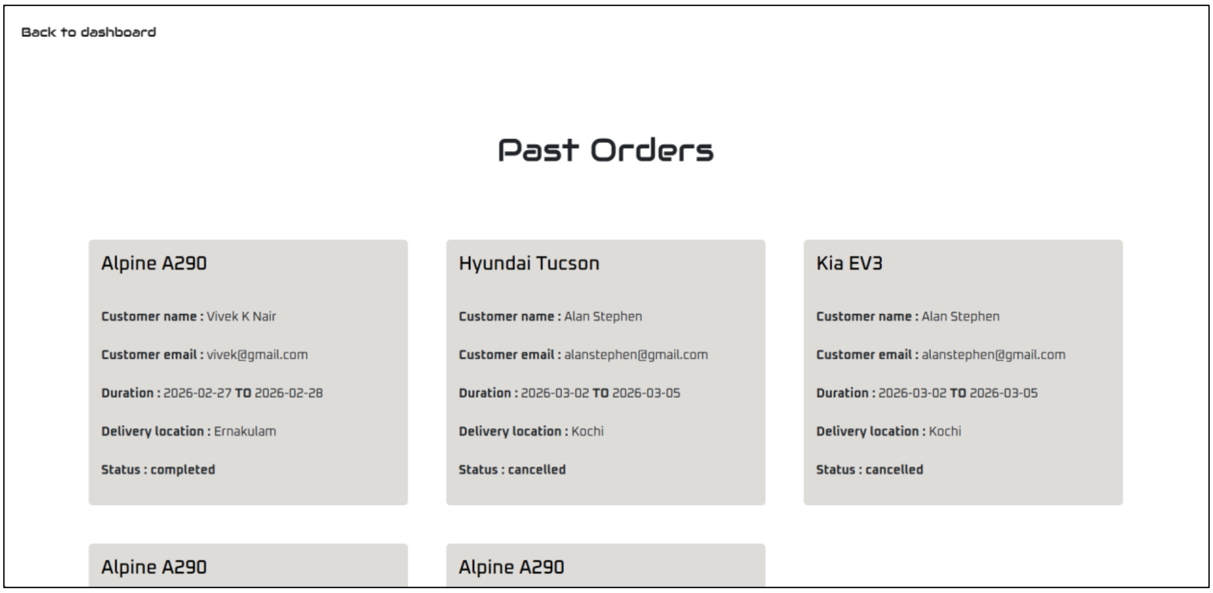

Rental Invoice
Invoice : #200R
Customer name : Alan Stephen
Email : alanstephen@gmail.com
Vehicle : Alpine A290
Duration : 2026-02-23 to 2026-02-27
Delivery location : Kochi
Status : completed
Total price : 10000.00
Print Invoice

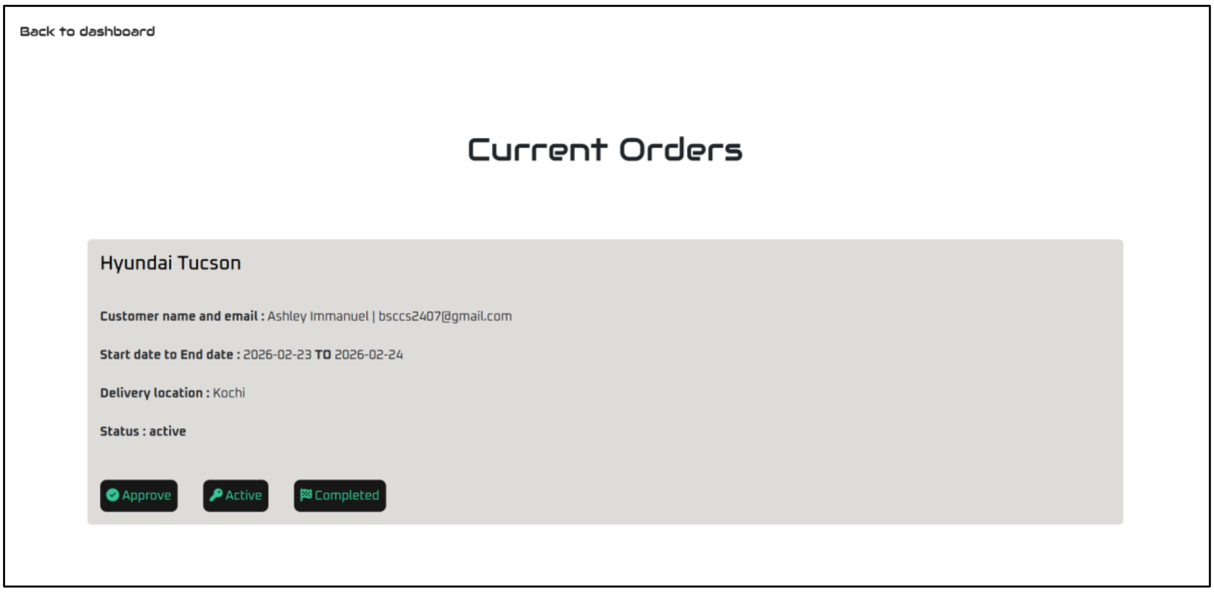
Admin Dashboard

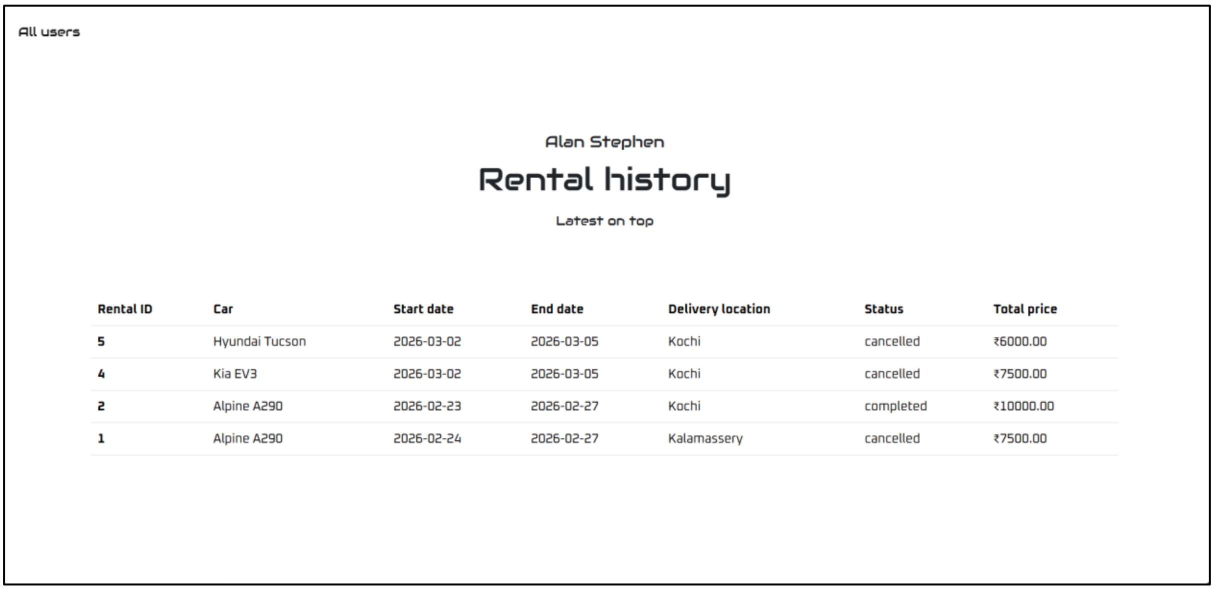












Rental ID	Car	Start date	End date	Delivery location	Status	Total price
5	Hyundai Tucson	2026-03-02	2026-03-05	Kochi	cancelled	₹6000.00
4	Kia EV3	2026-03-02	2026-03-05	Kochi	cancelled	₹7500.00
2	Alpine A290	2026-02-23	2026-02-27	Kochi	completed	₹10000.00
1	Alpine A290	2026-02-24	2026-02-27	Kalamassery	cancelled	₹7500.00